Validation supplement: quantile and event-probability compatibility

Rates and rate CIs are percentages to one decimal; rate differences are percentage points. Regression coefficients and their CIs use three significant figures. Brier means use the five-decimal source-table precision; other continuous error/width/distance summaries use three significant figures. Counts and protocol identifiers are unchanged. Full-precision source CSVs accompany this document. NA denotes an empty or unidentifiable estimate and is not zero. Engine counts are explicit; repeated fixed fits are not additional independent engines. In incompatible/compatible group counts, the same engine can occur in both groups across fixed fits, so these counts can overlap. S-table labels below are fixed identifiers.

S0. Locked rule and numerical addendum

Association protocol SHA256: `da781da5c725624ab53503d65cff89037e5ce15f0769a23b27e6d6a04eeb4540`.

> Protocol text (verbatim): require both real outcomes and both synthetic deviations in both families. The same at least six of eight real groups must have positive flag coefficients with 95% engine-bootstrap CIs excluding zero for BOTH median squared error and Brier after the specified RUL adjustment. Synthetic confirmation requires positive adjusted flag AND distance coefficients, with 95% CIs excluding zero, for BOTH squared oracle probability deviation and mean absolute oracle quantile deviation in BOTH families. Family-level effects are the equally weighted mean of the eight temporal subset/architecture coefficients or four covariate subset coefficients, using weighted-raw, final-window, 10,000-engine settings only. Bootstrap draws share engine resamples within each subset across architectures/variants, independently across subsets. Every individual setting is also reported; no choice of favorable settings. Rank-deficient required components fail the rule. If either real or synthetic condition fails, report “not informative” under this rule. No subgroup search or post-result revision.

Numerical implementation correction, 2026-11-01

The initial run screened the entire covariance matrix for full rank. Inspection showed only two distinct oracle pairs in the FD001 final-window fleet. Cubic nuisance columns are therefore redundant; this alone does not make the requested flag/distance coefficient unidentifiable. The initial partial outputs and run log are retained under initial_rank_screen. That run was interrupted after discovering this issue; no fit was removed.

Before rerunning: preserve the identical regression column space, use the Moore-Penrose solution for redundant nuisance columns, and check target identifiability using the target row of I-A^+A. Return NA only when the target is unidentifiable. SVD relative tolerance 1e-10; target null-space row norm tolerance 1e-6. Report full matrix rank, target identifiability, and condition diagnostics in a separate ledger. Use the same correction for all settings. This changes numerical handling, not outcome definitions, covariates, resampling, cohorts, decision thresholds or hypothesis. The original requirement to report rank-deficient point regressions as NA is retained for genuinely unidentifiable target effects; redundant nuisance parameterizations are disclosed. Both initial nominal-rank-screen outputs and corrected estimable-target results remain available.

Previously observed real branch remains 0/8 jointly passing groups. The corrected run is not a new prospectively unseen experiment; it is a documented implementation correction after a rank-screening issue, with all original outputs retained.

S0a. Known-law pipeline check: design and limits

The underlying Weibull shape is 1: the exponential excess lifetime is memoryless. The covariate is binary, x in {-1,+1}, with scale(x)=60 exp(0.5x). For final-window evaluation, true conditional RUL depends on x but is independent of age given x; the first feature directly supplies x, while other channels include age and noise. This simplified conditional law and feature construction are not a C-MAPSS-matched pipeline. Matching empirical truncation-age marginals does not match the real conditional RUL/history structure.

Per-fit incompatibility rates occur near 0.0% or 100.0% in the recorded pipeline-check distribution. Among non-collapsed fits, FD001 LSTM has median 99.7% and 97.5th percentile 100.0% (22 fits, 60 fitting engines, 10000 evaluation engines). FD003 LSTM also has 97.5th percentile 100.0% (28 fits, 60 fitting engines, 10000 evaluation engines). FD001 hybrid’s real rate of 13.6% on 100 test engines is below the pipeline-check 2.5th percentile of 14.3% from 5 non-collapsed fits evaluated on 10000 engines after fitting on 60 engines. The full min/2.5/25/50/75/97.5/max table is Table S11.

An upper percentile of 100.0% cannot be exceeded by an incompatibility rate. Such a comparison supplies no validation of the real rates. This degenerate known-law pipeline check neither confirms nor refutes the real-data incompatibility rates. The oracle-association results in Table S8 come from this same design and are descriptive only.

Archived pipeline construction and numerical checks

The synthetic law uses Weibull lifetimes with a simplified covariate relationship and a known conditional RUL distribution. Final-window sampling matches observed C-MAPSS truncation-length marginals; uniform-endpoint evaluation supplies a separate sampling policy. Oracle quantities are evaluated analytically or by deterministic inversion. Temporal models retain independently estimated quantile and event heads and the recorded training objective and budget. This is a known-law pipeline check, including approximation and optimization effects. Covariate-only networks and separately estimated parametric controls provide additional fitted controls within this design. Weighted raw outputs, TrainOff-corrected outputs and unweighted objectives remain separate arms.

Fitting fleets contain 60/156/60/149 engines for FD001–FD004. Evaluation uses both 100/259/100/248-engine fleets and 10,000-engine fleets per subset. The temporal cohorts retain 30 fits per architecture/subset/objective; covariate and parametric controls retain 200 per setting. Tables S1a–S1d report all-fit, non-collapsed and collapsed final-window weighted-raw references; the full policy/objective/size summary accompanies them.

For the FD001 final-window oracle check (Table S2), true quantiles with the true probability give incompatibility rate 0.0% and mean distance 0.0 on 10000 engines. The independently trained temporal LSTM weighted-raw rate is 67.5% on 10000 evaluation engines after fitting on 60 engines. These values describe this fitted pipeline only.

Archived oracle associations and original decision components

The following material is retained solely as a descriptive record of the same degenerate known-law design and the original locked rule. It is not support for the real-model claim.

Oracle deviations are squared probability error against the known conditional event probability and mean absolute quantile error against the known conditional quantiles. These oracle associations come from the same degenerate shape-1, binary-covariate design described in Supplement S0a. They are reported descriptively and are not support for the real-model central claim. Within this pipeline check, both temporal oracle deviations have positive adjusted flag and distance associations, whereas the covariate family gives mixed directions (10000 evaluation engines per subset; fitting engines 60/156/60/149). For oracle quantile MAE, the temporal family flag coefficient is 1.21 cycles, CI [1.19, 1.24]; the covariate coefficient is -1.89 cycles, CI [-1.91, -1.86]. These are equally weighted family summaries, not independent-model samples. The remaining coefficients are listed below.

- Temporal, squared-probability flag: coefficient 0.00501; CI [0.00492, 0.00511].
- Temporal, squared-probability distance: coefficient 0.00141; CI [0.00138, 0.00145].
- Temporal, quantile-MAE distance: coefficient 0.428; CI [0.420, 0.437].
- Covariate, squared-probability flag: coefficient -0.00124; CI [-0.00151, -0.000952].
- Covariate, squared-probability distance: coefficient 0.000923; CI [0.000811, 0.00104].
- Covariate, quantile-MAE distance: coefficient -0.655; CI [-0.670, -0.640].

Distance effects are per 0.1 probability-distance unit; all brackets are 95% engine-bootstrap CIs, with 10000 evaluation engines per subset and fitting counts 60/156/60/149. Tables S7a–S7b give the complete real associations and RUL-bin contrasts; Table S8 gives all primary oracle coefficients and links the full setting-specific association CSV.

Table S0a-1. Original main-text decision-component table

Component	Saved result	Evaluation engines
Real positive median-error associations	0/8 groups	100/259/100/248 per subset
Real positive Brier associations	6/8 groups	100/259/100/248 per subset
Real groups passing both outcomes	0/8 groups	100/259/100/248 per subset
Temporal required synthetic coefficients	4/4 pass	10000 per subset
Covariate required synthetic coefficients	1/4 pass	10000 per subset

Saved components of the original locked decision rule. Positive means that the corresponding 95% CI excludes zero in the positive direction; the joint verdict remains not informative. Synthetic components are protocol records, not evidence validating the real-data rates.

S0b. Synthetic gap and optimization checks

Gap collapse is defined by a median lower softplus gap below 1.25 cycles. Synthetic strata use a separate uniform-endpoint audit fleet of 10000 engines; real classification uses calibration engines. The retained real fits show no collapse under this definition on 20/52/20/50 calibration engines for FD001–FD004. The initialization-only intervention removes collapse in its tested synthetic cohort, yet incompatibility remains (60 fitting engines and 10000 evaluation engines). Tables S4a–S4c and S5 give the initialization/loss-scale/learning-rate/clipping outcomes; the frozen paper models are unchanged.

S0c. AI-use model and build details

OpenAI Codex assisted with code implementation and correction, analysis scripting, figure preparation, language editing and manuscript revision. The recorded configured model is `gpt-6.1-sol`, runtime `0.166.0-alpha.5.1`, and desktop build `26.1027.3384.0`. The runtime executable SHA256 is `3097c909fb6d5cd63510d44aceb1d42df9f6e85722a3e447e477def1ecf60e16`. The exact served checkpoint is unavailable. These identifiers are copied from the saved AI provenance records; they are not inferred from the product name. The author remains responsible for checking the text, evidence and submitted version.

Revision date: 2026-11-01. The exact served checkpoint remains unavailable.

S0d. Full chronological analysis history

The table records every identifiable stage from the skeptical-review findings. “After results” means that the stage was chosen after earlier results were examined. Initial choice history is unresolved; local locks do not establish independent registration or an untouched scientific design.

Analysis stage, in chronological order	Chosen after seeing earlier results?
Initial benchmark, architecture/loss, event horizon, RUL cap and compatibility choices; equal-search fitting and selection	Initial choice history unknown; equal-search design followed earlier benchmark development, although fitting selection used validation error.
Real-table reproduction gate and synthetic/quantile-perturbation validation, including final temporal synthetic design	Yes; real-model incompatibility rates had already been examined.
Extreme-null investigation of near-coincident quantiles	Yes; followed extreme synthetic incompatibility and small gaps.
Collapse cutoff and collapse-conditioned null analysis	Yes; followed the extreme-null investigation.
Factorial collapse-cause experiment	Yes; followed inspected collapse results.
Fixed initialization intervention	Yes; followed inspected collapse results.
Real-model weighting attribution and unweighted retraining comparisons	Yes; chosen in the collapse/weighting follow-up.
RUL-adjusted median-error/Brier associations, RUL bins, incremental probability adjustment and threshold-band rates	Yes; followed incompatibility/collapse results; the band was fixed before the first association calculation.
Additional probability-plus-band flag adjustment and operational decision rule	Yes; followed the association results.
Synthetic percentile/difficulty comparisons in the retained non-collapsed stratum	Yes; followed earlier synthetic results.
Event-head replacements and logit shifts; direct probability projection	Yes; followed compatibility/association results.
Own-mean effect ratios, MDEs and pooled effects	Yes; part of the later controls/projection/power analysis.
Expanded-model standard-calibrator comparison and calibrator-then-projection paths	Yes; followed the earlier controls/projection/association results.
Expanded-roster projection consistency counts and fixed distance-bin analysis	Yes; chosen for the expanded-model follow-up.
Projection mechanism/base-rate checks and replicated adjusted associations, including the restricted threshold band	Yes; chosen for the expanded-model follow-up.

Table S1a. Known-law pipeline-check strata: final-window, weighted raw, 10000 engines

Source: `null_stratified_summary.csv`. Rows: 48. The full source is in `validation_data`.

Family / group	Fit / test engines	Stratum	Fits	Rate [95% CI]	Mean distance
temporal FD001 lstm	60/10000	all	30	67.5% [67.5%, 67.6%]	0.150
temporal FD001 lstm	60/10000	noncollapsed	22	60.3% [60.2%, 60.3%]	0.0782
temporal FD001 lstm	60/10000	collapsed	8	87.5% [87.5%, 87.5%]	0.348
temporal FD001 hybrid	60/10000	all	30	65.7% [65.3%, 66.0%]	0.149
temporal FD001 hybrid	60/10000	noncollapsed	5	42.6% [42.5%, 42.8%]	0.0310
temporal FD001 hybrid	60/10000	collapsed	25	70.3% [69.9%, 70.7%]	0.172
covariate FD001 covariate	60/10000	all	200	34.1% [33.5%, 34.8%]	0.0519
covariate FD001 covariate	60/10000	noncollapsed	200	34.1% [33.5%, 34.8%]	0.0519
covariate FD001 covariate	60/10000	collapsed	0	NA [NA, NA]	NA
parametric FD001 independent exact family	60/10000	all	200	13.7% [13.4%, 14.0%]	0.0146
parametric FD001 independent exact family	60/10000	noncollapsed	200	13.7% [13.4%, 14.0%]	0.0146
parametric FD001 independent exact family	60/10000	collapsed	0	NA [NA, NA]	NA
temporal FD002 lstm	156/10000	all	30	26.8% [26.3%, 27.2%]	0.0407
temporal FD002 lstm	156/10000	noncollapsed	30	26.8% [26.3%, 27.2%]	0.0407
temporal FD002 lstm	156/10000	collapsed	0	NA [NA, NA]	NA
temporal FD002 hybrid	156/10000	all	30	45.0% [44.8%, 45.2%]	0.0873
temporal FD002 hybrid	156/10000	noncollapsed	10	25.7% [25.4%, 26.0%]	0.0139
temporal FD002 hybrid	156/10000	collapsed	20	54.6% [54.2%, 55.1%]	0.124
covariate FD002 covariate	156/10000	all	200	12.2% [12.0%, 12.4%]	0.0188
covariate FD002 covariate	156/10000	noncollapsed	200	12.2% [12.0%, 12.4%]	0.0188
covariate FD002 covariate	156/10000	collapsed	0	NA [NA, NA]	NA
parametric FD002 independent exact family	156/10000	all	200	7.3% [7.1%, 7.4%]	0.00783
parametric FD002 independent exact family	156/10000	noncollapsed	200	7.3% [7.1%, 7.4%]	0.00783
parametric FD002 independent exact family	156/10000	collapsed	0	NA [NA, NA]	NA
temporal FD003 lstm	60/10000	all	30	42.9% [42.6%, 43.2%]	0.0337
temporal FD003 lstm	60/10000	noncollapsed	28	40.6% [40.2%, 41.0%]	0.0348
temporal FD003 lstm	60/10000	collapsed	2	75.1% [74.5%, 75.6%]	0.0186
temporal FD003 hybrid	60/10000	all	30	47.1% [46.9%, 47.3%]	0.0843

Family / group	Fit / test engines	Stratum	Fits	Rate [95% CI]	Mean distance
temporal FD003 hybrid	60/10000	noncollapsed	17	34.3% [33.8%, 34.7%]	0.0721
temporal FD003 hybrid	60/10000	collapsed	13	63.9% [63.6%, 64.2%]	0.100
covariate FD003 covariate	60/10000	all	200	37.1% [36.5%, 37.7%]	0.0550
covariate FD003 covariate	60/10000	noncollapsed	200	37.1% [36.5%, 37.7%]	0.0550
covariate FD003 covariate	60/10000	collapsed	0	NA [NA, NA]	NA
parametric FD003 independent exact family	60/10000	all	200	13.7% [13.5%, 14.0%]	0.0146
parametric FD003 independent exact family	60/10000	noncollapsed	200	13.7% [13.5%, 14.0%]	0.0146
parametric FD003 independent exact family	60/10000	collapsed	0	NA [NA, NA]	NA
temporal FD004 lstm	149/10000	all	30	29.4% [28.9%, 29.9%]	0.0465
temporal FD004 lstm	149/10000	noncollapsed	30	29.4% [28.9%, 29.9%]	0.0465
temporal FD004 lstm	149/10000	collapsed	0	NA [NA, NA]	NA
temporal FD004 hybrid	149/10000	all	30	44.7% [44.7%, 44.8%]	0.0521
temporal FD004 hybrid	149/10000	noncollapsed	18	32.6% [32.3%, 32.8%]	0.0252
temporal FD004 hybrid	149/10000	collapsed	12	63.0% [62.8%, 63.3%]	0.0925
covariate FD004 covariate	149/10000	all	200	14.4% [14.2%, 14.7%]	0.0214
covariate FD004 covariate	149/10000	noncollapsed	200	14.4% [14.2%, 14.7%]	0.0214
covariate FD004 covariate	149/10000	collapsed	0	NA [NA, NA]	NA
parametric FD004 independent exact family	149/10000	all	200	5.8% [5.7%, 5.9%]	0.00587
parametric FD004 independent exact family	149/10000	noncollapsed	200	5.8% [5.7%, 5.9%]	0.00587
parametric FD004 independent exact family	149/10000	collapsed	0	NA [NA, NA]	NA

115 ## Table S1b. Temporal cross-pipeline numerical comparisons, all strata; not a benchmark

116 Source: `comparison_temporal.csv`. Rows: 24. The full source is in `validation_data`.

Real group	Fit / null / real engines	Stratum	Pipeline fits	Real rate	Pipeline 97.5th	Exceeds
FD001 LSTM	60/10000/100	all	30	20.4%	100.0%	False
FD001 LSTM	60/10000/100	noncollapsed	22	20.4%	100.0%	False
FD001 LSTM	60/10000/100	collapsed	8	20.4%	100.0%	False
FD001 Hybrid	60/10000/100	all	30	13.6%	100.0%	False
FD001 Hybrid	60/10000/100	noncollapsed	5	13.6%	55.2%	False
FD001 Hybrid	60/10000/100	collapsed	25	13.6%	100.0%	False
FD002 LSTM	156/10000/259	all	30	9.6%	57.5%	False
FD002 LSTM	156/10000/259	noncollapsed	30	9.6%	57.5%	False

Real group	Fit / null / real engines	Stratum	Pipeline fits	Real rate	Pipeline 97.5th	Exceeds
FD002 LSTM	156/10000/259	collapsed	0	9.6%	NA	NA
FD002 Hybrid	156/10000/259	all	30	8.9%	100.0%	False
FD002 Hybrid	156/10000/259	noncollapsed	10	8.9%	88.8%	False
FD002 Hybrid	156/10000/259	collapsed	20	8.9%	100.0%	False
FD003 LSTM	60/10000/100	all	30	12.8%	100.0%	False
FD003 LSTM	60/10000/100	noncollapsed	28	12.8%	100.0%	False
FD003 LSTM	60/10000/100	collapsed	2	12.8%	96.5%	False
FD003 Hybrid	60/10000/100	all	30	11.8%	97.9%	False
FD003 Hybrid	60/10000/100	noncollapsed	17	11.8%	79.3%	False
FD003 Hybrid	60/10000/100	collapsed	13	11.8%	99.1%	False
FD004 LSTM	149/10000/248	all	30	16.6%	60.2%	False
FD004 LSTM	149/10000/248	noncollapsed	30	16.6%	60.2%	False
FD004 LSTM	149/10000/248	collapsed	0	16.6%	NA	NA
FD004 Hybrid	149/10000/248	all	30	12.7%	96.8%	False
FD004 Hybrid	149/10000/248	noncollapsed	18	12.7%	94.5%	False
FD004 Hybrid	149/10000/248	collapsed	12	12.7%	97.9%	False

Table S1c. Covariate cross-pipeline numerical comparisons, all strata; not a benchmark

Source: `comparison_covariate.csv`. Rows: 24. The full source is in `validation_data`.

Real group	Fit / null / real engines	Stratum	Pipeline fits	Real rate	Pipeline 97.5th	Exceeds
FD001 LSTM	60/10000/100	all	200	20.4%	60.6%	False
FD001 Hybrid	60/10000/100	all	200	13.6%	60.6%	False
FD001 LSTM	60/10000/100	noncollapsed	200	20.4%	60.6%	False
FD001 Hybrid	60/10000/100	noncollapsed	200	13.6%	60.6%	False
FD001 LSTM	60/10000/100	collapsed	0	20.4%	NA	NA
FD001 Hybrid	60/10000/100	collapsed	0	13.6%	NA	NA
FD002 LSTM	156/10000/259	all	200	9.6%	50.3%	False
FD002 Hybrid	156/10000/259	all	200	8.9%	50.3%	False
FD002 LSTM	156/10000/259	noncollapsed	200	9.6%	50.3%	False
FD002 Hybrid	156/10000/259	noncollapsed	200	8.9%	50.3%	False
FD002 LSTM	156/10000/259	collapsed	0	9.6%	NA	NA
FD002 Hybrid	156/10000/259	collapsed	0	8.9%	NA	NA
FD003 LSTM	60/10000/100	all	200	12.8%	69.3%	False
FD003 Hybrid	60/10000/100	all	200	11.8%	69.3%	False
FD003 LSTM	60/10000/100	noncollapsed	200	12.8%	69.3%	False
FD003 Hybrid	60/10000/100	noncollapsed	200	11.8%	69.3%	False
FD003 LSTM	60/10000/100	collapsed	0	12.8%	NA	NA
FD003 Hybrid	60/10000/100	collapsed	0	11.8%	NA	NA
FD004 LSTM	149/10000/248	all	200	16.6%	50.3%	False
FD004 Hybrid	149/10000/248	all	200	12.7%	50.3%	False
FD004 LSTM	149/10000/248	noncollapsed	200	16.6%	50.3%	False
FD004 Hybrid	149/10000/248	noncollapsed	200	12.7%	50.3%	False
FD004 LSTM	149/10000/248	collapsed	0	16.6%	NA	NA
FD004 Hybrid	149/10000/248	collapsed	0	12.7%	NA	NA

Table S1d. Parametric cross-pipeline numerical comparisons, all strata; not a benchmark

Source: `comparison_parametric.csv`. Rows: 24. The full source is in `validation_data`.

Real group	Fit / null / real engines	Stratum	Pipeline fits	Real rate	Pipeline 97.5th	Exceeds
FD001 LSTM	60/10000/100	all	200	20.4%	49.8%	False

Real group	Fit / null / real engines	Stratum	Pipeline fits	Real rate	Pipeline 97.5th	Exceeds
FD001 Hybrid	60/10000/100	all	200	13.6%	49.8%	False
FD001 LSTM	60/10000/100	noncollapsed	200	20.4%	49.8%	False
FD001 Hybrid	60/10000/100	noncollapsed	200	13.6%	49.8%	False
FD001 LSTM	60/10000/100	collapsed	0	20.4%	NA	NA
FD001 Hybrid	60/10000/100	collapsed	0	13.6%	NA	NA
FD002 LSTM	156/10000/259	all	200	9.6%	50.1%	False
FD002 Hybrid	156/10000/259	all	200	8.9%	50.1%	False
FD002 LSTM	156/10000/259	noncollapsed	200	9.6%	50.1%	False
FD002 Hybrid	156/10000/259	noncollapsed	200	8.9%	50.1%	False
FD002 LSTM	156/10000/259	collapsed	0	9.6%	NA	NA
FD002 Hybrid	156/10000/259	collapsed	0	8.9%	NA	NA
FD003 LSTM	60/10000/100	all	200	12.8%	49.9%	False
FD003 Hybrid	60/10000/100	all	200	11.8%	49.9%	False
FD003 LSTM	60/10000/100	noncollapsed	200	12.8%	49.9%	False
FD003 Hybrid	60/10000/100	noncollapsed	200	11.8%	49.9%	False
FD003 LSTM	60/10000/100	collapsed	0	12.8%	NA	NA
FD003 Hybrid	60/10000/100	collapsed	0	11.8%	NA	NA
FD004 LSTM	149/10000/248	all	200	16.6%	50.2%	False
FD004 Hybrid	149/10000/248	all	200	12.7%	50.2%	False
FD004 LSTM	149/10000/248	noncollapsed	200	16.6%	50.2%	False
FD004 Hybrid	149/10000/248	noncollapsed	200	12.7%	50.2%	False
FD004 LSTM	149/10000/248	collapsed	0	16.6%	NA	NA
FD004 Hybrid	149/10000/248	collapsed	0	12.7%	NA	NA

Table S2. Oracle quantiles with true probability and controlled quantile noise

Source: `synthetic_oracle_noise.csv`. Rows: 64. The full source is in `validation_data`.

Subset / policy	Fit / test engines	Noise	Rate [95% CI]	Mean distance
FD001 final	0/100	0.0%	0.0% [0.0%, 0.0%]	0.00
FD001 final	0/10000	0.0%	0.0% [0.0%, 0.0%]	0.00
FD001 final	0/100	5.0%	0.0% [0.0%, 0.0%]	0.00
FD001 final	0/10000	5.0%	0.0% [0.0%, 0.0%]	0.00000212
FD001 final	0/100	10.0%	1.0% [0.7%, 1.2%]	0.00261
FD001 final	0/10000	10.0%	1.2% [1.1%, 1.2%]	0.00308
FD001 final	0/100	20.0%	11.6% [10.9%, 12.4%]	0.0401
FD001 final	0/10000	20.0%	12.2% [12.1%, 12.3%]	0.0401
FD001 uniform	0/100	0.0%	0.0% [0.0%, 0.0%]	0.00
FD001 uniform	0/10000	0.0%	0.0% [0.0%, 0.0%]	0.00
FD001 uniform	0/100	5.0%	0.7% [0.0%, 1.9%]	0.00000950
FD001 uniform	0/10000	5.0%	0.2% [0.2%, 0.2%]	0.00000500
FD001 uniform	0/100	10.0%	1.6% [0.6%, 3.0%]	0.00287
FD001 uniform	0/10000	10.0%	0.9% [0.9%, 1.0%]	0.00242
FD001 uniform	0/100	20.0%	9.4% [7.9%, 11.1%]	0.0364
FD001 uniform	0/10000	20.0%	8.1% [8.0%, 8.3%]	0.0334
FD002 final	0/259	0.0%	0.0% [0.0%, 0.0%]	0.00
FD002 final	0/10000	0.0%	0.0% [0.0%, 0.0%]	0.00
FD002 final	0/259	5.0%	0.0% [0.0%, 0.0%]	0.00
FD002 final	0/10000	5.0%	0.0% [0.0%, 0.0%]	0.00000148
FD002 final	0/259	10.0%	1.1% [1.0%, 1.3%]	0.00307
FD002 final	0/10000	10.0%	1.1% [1.1%, 1.2%]	0.00309
FD002 final	0/259	20.0%	12.4% [11.8%, 12.9%]	0.0403
FD002 final	0/10000	20.0%	12.1% [12.1%, 12.2%]	0.0399
FD002 uniform	0/259	0.0%	0.0% [0.0%, 0.0%]	0.00
FD002 uniform	0/10000	0.0%	0.0% [0.0%, 0.0%]	0.00
FD002 uniform	0/259	5.0%	0.2% [0.1%, 0.5%]	0.0000230

Subset / policy	Fit / test engines	Noise	Rate [95% CI]	Mean distance
FD002 uniform	0/10000	5.0%	0.2% [0.1%, 0.2%]	0.00000587
FD002 uniform	0/259	10.0%	1.2% [0.8%, 1.8%]	0.00289
FD002 uniform	0/10000	10.0%	1.0% [0.9%, 1.0%]	0.00236
FD002 uniform	0/259	20.0%	8.5% [7.7%, 9.4%]	0.0341
FD002 uniform	0/10000	20.0%	8.1% [8.0%, 8.3%]	0.0332
FD003 final	0/100	0.0%	0.0% [0.0%, 0.0%]	0.00
FD003 final	0/10000	0.0%	0.0% [0.0%, 0.0%]	0.00
FD003 final	0/100	5.0%	0.0% [0.0%, 0.0%]	0.0000212
FD003 final	0/10000	5.0%	0.0% [0.0%, 0.0%]	0.00000127
FD003 final	0/100	10.0%	1.2% [1.0%, 1.5%]	0.00337
FD003 final	0/10000	10.0%	1.2% [1.1%, 1.2%]	0.00310
FD003 final	0/100	20.0%	12.5% [11.7%, 13.3%]	0.0419
FD003 final	0/10000	20.0%	12.2% [12.2%, 12.3%]	0.0402
FD003 uniform	0/100	0.0%	0.0% [0.0%, 0.0%]	0.00
FD003 uniform	0/10000	0.0%	0.0% [0.0%, 0.0%]	0.00
FD003 uniform	0/100	5.0%	0.0% [0.0%, 0.0%]	6.52e-7
FD003 uniform	0/10000	5.0%	0.2% [0.1%, 0.2%]	0.00000567
FD003 uniform	0/100	10.0%	0.8% [0.6%, 1.2%]	0.00262
FD003 uniform	0/10000	10.0%	1.0% [0.9%, 1.0%]	0.00243
FD003 uniform	0/100	20.0%	8.5% [7.2%, 9.9%]	0.0365
FD003 uniform	0/10000	20.0%	8.2% [8.1%, 8.3%]	0.0335
FD004 final	0/248	0.0%	0.0% [0.0%, 0.0%]	0.00
FD004 final	0/10000	0.0%	0.0% [0.0%, 0.0%]	0.00
FD004 final	0/248	5.0%	0.0% [0.0%, 0.0%]	0.00
FD004 final	0/10000	5.0%	0.0% [0.0%, 0.0%]	0.00000233
FD004 final	0/248	10.0%	1.1% [1.0%, 1.3%]	0.00290
FD004 final	0/10000	10.0%	1.2% [1.1%, 1.2%]	0.00310
FD004 final	0/248	20.0%	12.5% [12.0%, 13.0%]	0.0414
FD004 final	0/10000	20.0%	12.2% [12.1%, 12.3%]	0.0402
FD004 uniform	0/248	0.0%	0.0% [0.0%, 0.0%]	0.00
FD004 uniform	0/10000	0.0%	0.0% [0.0%, 0.0%]	0.00
FD004 uniform	0/248	5.0%	0.0% [0.0%, 0.0%]	0.00
FD004 uniform	0/10000	5.0%	0.2% [0.1%, 0.2%]	0.00000481
FD004 uniform	0/248	10.0%	0.5% [0.4%, 0.7%]	0.00225
FD004 uniform	0/10000	10.0%	0.9% [0.9%, 1.0%]	0.00239
FD004 uniform	0/248	20.0%	7.8% [7.1%, 8.5%]	0.0343
FD004 uniform	0/10000	20.0%	8.2% [8.1%, 8.4%]	0.0337

Table S3. Perturbation sensitivity, all groups and noise levels

Source: `perturbation_summary.csv`. Rows: 32. The full source is in `validation_data`.

Group	Test engines	Noise	Rate [95% CI]	Change, pp [95% CI]	Mean distance	Crossing rate
FD001 lstm	100	0.0%	20.4% [14.4%, 26.8%]	0.0 [0.0, 0.0]	0.0210	0.0%
FD001 lstm	100	5.0%	20.5% [14.7%, 26.8%]	0.1 [-0.5, 0.7]	0.0217	0.0%
FD001 lstm	100	10.0%	20.4% [14.6%, 26.6%]	-0.0 [-0.9, 0.9]	0.0230	0.0%
FD001 lstm	100	20.0%	21.0% [15.4%, 27.1%]	0.6 [-0.8, 2.0]	0.0276	3.6%
FD001 hybrid	100	0.0%	13.6% [8.8%, 19.2%]	0.0 [0.0, 0.0]	0.0138	0.0%
FD001 hybrid	100	5.0%	13.1% [8.4%, 18.4%]	-0.5 [-1.1, -0.0]	0.0133	0.0%
FD001 hybrid	100	10.0%	13.0% [8.4%, 18.2%]	-0.6 [-1.6, 0.2]	0.0134	0.0%

Group	Test engines	Noise	Rate [95% CI]	Change, pp [95% CI]	Mean distance	Crossing rate
FD001 hybrid	100	20.0%	13.0% [8.7%, 18.0%]	-0.6 [-2.0, 0.7]	0.0149	2.1%
FD002 lstm	259	0.0%	9.6% [6.9%, 12.7%]	0.0 [0.0, 0.0]	0.00671	0.0%
FD002 lstm	259	5.0%	9.9% [7.3%, 12.9%]	0.3 [-0.0, 0.8]	0.00711	0.0%
FD002 lstm	259	10.0%	10.4% [7.7%, 13.3%]	0.8 [0.2, 1.4]	0.00783	0.0%
FD002 lstm	259	20.0%	12.0% [9.5%, 14.7%]	2.4 [1.4, 3.4]	0.0105	2.5%
FD002 hybrid	259	0.0%	8.9% [6.5%, 11.5%]	0.0 [0.0, 0.0]	0.00720	0.0%
FD002 hybrid	259	5.0%	8.9% [6.7%, 11.5%]	0.1 [-0.3, 0.4]	0.00773	0.0%
FD002 hybrid	259	10.0%	9.5% [7.3%, 12.0%]	0.7 [0.1, 1.2]	0.00882	0.0%
FD002 hybrid	259	20.0%	11.4% [9.1%, 13.9%]	2.5 [1.7, 3.4]	0.0119	3.0%
FD003 lstm	100	0.0%	12.8% [8.4%, 17.8%]	0.0 [0.0, 0.0]	0.0150	0.0%
FD003 lstm	100	5.0%	12.9% [8.5%, 17.8%]	0.1 [-0.5, 0.8]	0.0156	0.0%
FD003 lstm	100	10.0%	13.3% [8.9%, 18.2%]	0.5 [-0.4, 1.7]	0.0167	0.0%
FD003 lstm	100	20.0%	14.6% [10.2%, 19.3%]	1.8 [0.4, 3.4]	0.0196	1.7%
FD003 hybrid	100	0.0%	11.8% [8.0%, 16.4%]	0.0 [0.0, 0.0]	0.0149	0.0%
FD003 hybrid	100	5.0%	11.7% [7.9%, 16.1%]	-0.1 [-0.7, 0.4]	0.0151	0.0%
FD003 hybrid	100	10.0%	11.4% [7.7%, 15.8%]	-0.4 [-1.0, 0.2]	0.0155	0.0%
FD003 hybrid	100	20.0%	11.9% [8.2%, 16.2%]	0.1 [-0.7, 0.8]	0.0168	1.4%
FD004 lstm	248	0.0%	16.6% [13.0%, 20.4%]	0.0 [0.0, 0.0]	0.0137	0.0%
FD004 lstm	248	5.0%	16.6% [13.1%, 20.4%]	0.0 [-0.4, 0.4]	0.0143	0.0%
FD004 lstm	248	10.0%	17.0% [13.5%, 20.8%]	0.4 [-0.2, 1.0]	0.0162	0.1%
FD004 lstm	248	20.0%	18.5% [15.2%, 22.1%]	1.9 [0.9, 2.9]	0.0215	3.4%
FD004 hybrid	248	0.0%	12.7% [9.8%, 15.8%]	0.0 [0.0, 0.0]	0.00857	0.0%
FD004 hybrid	248	5.0%	12.7% [9.9%, 15.8%]	0.0 [-0.3, 0.4]	0.00902	0.0%
FD004 hybrid	248	10.0%	13.2% [10.5%, 16.2%]	0.6 [0.0, 1.1]	0.0104	0.1%
FD004 hybrid	248	20.0%	14.9% [12.2%, 17.9%]	2.2 [1.4, 3.1]	0.0145	4.1%

Table S4a. Real calibration-only collapse classification

Source: `real_collapse_fractions.csv`. Rows: 8. The full source is in `validation_data`.

Group	Fit / cal / test engines	Fits	Collapsed fits	Collapsed fraction
FD001 lstm	60/20/100	5	0	0.0%
FD001 hybrid	60/20/100	5	0	0.0%
FD002 lstm	156/52/259	5	0	0.0%

Group	Fit / cal / test engines	Fits	Collapsed fits	Collapsed fraction
FD002 hybrid	156/52/259	5	0	0.0%
FD003 lstm	60/20/100	5	0	0.0%
FD003 hybrid	60/20/100	5	0	0.0%
FD004 lstm	149/50/248	5	0	0.0%
FD004 hybrid	149/50/248	5	0	0.0%

Table S4b. Initialization-only intervention safeguards

Source: `fix_success_checks.csv`. Rows: 4. The full source is in `validation_data`.

Architecture / objective	Fit / test engines	Completed / failed	Collapsed fits	Oracle MAE change [95% CI]
lstm weighted	60/10000	30/0	0	-14.5 [-14.5, -14.5]
lstm unweighted	60/10000	30/0	0	-14.6 [-14.6, -14.6]
hybrid weighted	60/10000	30/0	0	-12.6 [-12.7, -12.6]
hybrid unweighted	60/10000	30/0	0	-12.7 [-12.7, -12.6]

Table S4c. Initialization-only intervention: final-window all-fit rates

Source: `fixed_init_summary.csv`. Rows: 6. The full source is in `validation_data`.

Group / variant	Fit / test engines	Fits	Rate [95% CI]	Mean distance
FD001 lstm weighted raw	60/10000	30	40.0% [39.8%, 40.1%]	0.0121
FD001 lstm weighted corrected	60/10000	30	64.0% [63.7%, 64.4%]	0.0859
FD001 lstm unweighted	60/10000	30	61.5% [61.1%, 61.9%]	0.0722
FD001 hybrid weighted raw	60/10000	30	38.3% [38.1%, 38.4%]	0.0215
FD001 hybrid weighted corrected	60/10000	30	37.0% [36.4%, 37.7%]	0.0525
FD001 hybrid unweighted	60/10000	30	34.5% [34.0%, 35.1%]	0.0222

Table S5. All factorial main/interaction contrast summaries

Source: `factorial_contrast_summary.csv`. Rows: 90. The full source is in `validation_data`.

Architecture / factors	Metric	Fit / test engines	Paired fleets	Mean contrast	Min / max contrast
lstm clipping disabled	collapsed	60/10000	3	0.00	0.00/0.00
lstm clipping disabled	rate	60/10000	3	0.0655	0.0349/0.0923
lstm clipping disabled	distance	60/10000	3	-0.000820	-0.00393/0.00195
lstm initialization	collapsed	60/10000	3	-0.5	-1.0/0.00
lstm initialization	rate	60/10000	3	-0.418	-0.672/-0.241
lstm initialization	distance	60/10000	3	-0.153	-0.239/-0.0762
lstm initialization*clipping disabled	collapsed	60/10000	3	0.00	0.00/0.00
lstm initialization*clipping disabled	rate	60/10000	3	-0.00399	-0.141/0.0705
lstm initialization*clipping disabled	distance	60/10000	3	0.00149	-0.00676/0.0135

Architecture / factors	Metric	Fit / test engines	Paired fleets	Mean contrast	Min / max contrast
lstm initialization*lr tripled	collapsed	60/10000	3	0.167	0.00/0.5
lstm initialization*lr tripled	rate	60/10000	3	0.182	-0.176/0.499
lstm initialization*lr tripled	distance	60/10000	3	0.218	0.147/0.277
lstm initialization*lr tripled*clipping disabled	collapsed	60/10000	3	-0.333	-1.0/0.00
lstm initialization*lr tripled*clipping disabled	rate	60/10000	3	-0.227	-0.345/-0.0553
lstm initialization*lr tripled*clipping disabled	distance	60/10000	3	-0.00918	-0.0158/-0.00584
lstm initialization*pinball normalized	collapsed	60/10000	3	0.00	0.00/0.00
lstm initialization*pinball normalized	rate	60/10000	3	0.369	0.205/0.477
lstm initialization*pinball normalized	distance	60/10000	3	0.00205	-0.121/0.174
lstm initialization*pinball normalized*clipping disabled	collapsed	60/10000	3	0.00	0.00/0.00
lstm initialization*pinball normalized*clipping disabled	rate	60/10000	3	0.0209	-0.141/0.249
lstm initialization*pinball normalized*clipping disabled	distance	60/10000	3	0.00563	0.00449/0.00781
lstm initialization*pinball normalized*lr tripled	collapsed	60/10000	3	0.333	0.00/1.0
lstm initialization*pinball normalized*lr tripled	rate	60/10000	3	-0.311	-0.520/-0.0459
lstm initialization*pinball normalized*lr tripled	distance	60/10000	3	-0.162	-0.372/0.253
lstm initialization*pinball normalized*lr tripled*clipping disabled	collapsed	60/10000	3	0.667	0.00/2.0
lstm initialization*pinball normalized*lr tripled*clipping disabled	rate	60/10000	3	0.481	0.254/0.690
lstm initialization*pinball normalized*lr tripled*clipping disabled	distance	60/10000	3	0.00329	-0.00643/0.0118
lstm lr tripled	collapsed	60/10000	3	-0.0833	-0.25/0.00
lstm lr tripled	rate	60/10000	3	-0.470	-0.639/-0.300

Architecture / factors	Metric	Fit / test engines	Paired fleets	Mean contrast	Min / max contrast
lstm lr tripled	distance	60/10000	3	-0.137	-0.175/-0.0921
lstm lr tripled*clipping disabled	collapsed	60/10000	3	0.167	0.00/0.5
lstm lr tripled*clipping disabled	rate	60/10000	3	0.0213	-0.0584/0.139
lstm lr tripled*clipping disabled	distance	60/10000	3	0.00707	0.00290/0.0129
lstm pinball normalized	collapsed	60/10000	3	0.00	0.00/0.00
lstm pinball normalized	rate	60/10000	3	-0.00962	-0.129/0.109
lstm pinball normalized	distance	60/10000	3	0.0271	-0.0550/0.0781
lstm pinball normalized*clipping disabled	collapsed	60/10000	3	0.00	0.00/0.00
lstm pinball normalized*clipping disabled	rate	60/10000	3	-0.116	-0.185/-0.0341
lstm pinball normalized*clipping disabled	distance	60/10000	3	-0.000110	-0.00390/0.00249
lstm pinball normalized*lr tripled	collapsed	60/10000	3	-0.167	-0.5/0.00
lstm pinball normalized*lr tripled	rate	60/10000	3	-0.204	-0.441/0.0417
lstm pinball normalized*lr tripled	distance	60/10000	3	0.0271	-0.165/0.127
lstm pinball normalized*lr tripled* clipping disabled	collapsed	60/10000	3	-0.333	-1.0/0.00
lstm pinball normalized*lr tripled* clipping disabled	rate	60/10000	3	-0.0119	-0.257/0.117
lstm pinball normalized*lr tripled* clipping disabled	distance	60/10000	3	-0.00149	-0.00757/0.00889
hybrid clipping disabled	collapsed	60/10000	3	-0.0417	-0.125/0.00
hybrid clipping disabled	rate	60/10000	3	-0.0485	-0.146/0.0409
hybrid clipping disabled	distance	60/10000	3	-0.0230	-0.0563/-0.00228
hybrid initialization	collapsed	60/10000	3	-0.875	-1.0/-0.625
hybrid initialization	rate	60/10000	3	-0.429	-0.548/-0.351
hybrid initialization	distance	60/10000	3	-0.114	-0.159/-0.0652
hybrid initialization*clipping disabled	collapsed	60/10000	3	0.0833	0.00/0.25
hybrid initialization*clipping disabled	rate	60/10000	3	0.125	0.0817/0.210
hybrid initialization*clipping disabled	distance	60/10000	3	0.0441	0.00588/0.114

Architecture / factors	Metric	Fit / test engines	Paired fleets	Mean contrast	Min / max contrast
hybrid initialization*lr tripled	collapsed	60/10000	3	0.0833	0.00/0.25
hybrid initialization*lr tripled	rate	60/10000	3	-0.0295	-0.0833/0.0729
hybrid initialization*lr tripled	distance	60/10000	3	0.0130	-0.0175/0.0507
hybrid initialization*lr tripled*clipping disabled	collapsed	60/10000	3	-0.167	-0.5/0.00
hybrid initialization*lr tripled*clipping disabled	rate	60/10000	3	-0.0302	-0.331/0.138
hybrid initialization*lr tripled*clipping disabled	distance	60/10000	3	-0.0831	-0.202/0.00557
hybrid initialization*pinball normalized	collapsed	60/10000	3	-0.0833	-0.25/0.00
hybrid initialization*pinball normalized	rate	60/10000	3	0.129	-0.0529/0.278
hybrid initialization*pinball normalized	distance	60/10000	3	0.0345	0.0109/0.0581
hybrid initialization*pinball normalized*clipping disabled	collapsed	60/10000	3	-0.5	-1.5/0.00
hybrid initialization*pinball normalized*clipping disabled	rate	60/10000	3	-0.650	-1.51/-0.154
hybrid initialization*pinball normalized*clipping disabled	distance	60/10000	3	-0.170	-0.479/0.0184
hybrid initialization*pinball normalized*lr tripled	collapsed	60/10000	3	0.167	0.00/0.5
hybrid initialization*pinball normalized*lr tripled	rate	60/10000	3	-0.0159	-0.194/0.284
hybrid initialization*pinball normalized*lr tripled	distance	60/10000	3	-0.0386	-0.108/0.0940
hybrid initialization*pinball normalized*lr tripled*clipping disabled	collapsed	60/10000	3	-0.333	-1.0/0.00
hybrid initialization*pinball normalized*lr tripled*clipping disabled	rate	60/10000	3	-0.483	-0.548/-0.375

Architecture / factors	Metric	Fit / test engines	Paired fleets	Mean contrast	Min / max contrast
hybrid initialization*pinball normalized*lr tripled*clipping disabled	distance	60/10000	3	-0.132	-0.268/-0.0297
hybrid lr tripled	collapsed	60/10000	3	-0.0417	-0.125/0.00
hybrid lr tripled	rate	60/10000	3	-0.0254	-0.0417/-0.0101
hybrid lr tripled	distance	60/10000	3	-0.0129	-0.0263/0.00114
hybrid lr tripled*clipping disabled	collapsed	60/10000	3	0.0833	0.00/0.25
hybrid lr tripled*clipping disabled	rate	60/10000	3	0.0153	-0.109/0.0863
hybrid lr tripled*clipping disabled	distance	60/10000	3	0.0279	-0.0272/0.0981
hybrid pinball normalized	collapsed	60/10000	3	0.0417	0.00/0.125
hybrid pinball normalized	rate	60/10000	3	-0.0457	-0.359/0.149
hybrid pinball normalized	distance	60/10000	3	-0.000639	-0.00449/0.00292
hybrid pinball normalized*clipping disabled	collapsed	60/10000	3	0.25	0.00/0.75
hybrid pinball normalized*clipping disabled	rate	60/10000	3	0.210	-0.0769/0.691
hybrid pinball normalized*clipping disabled	distance	60/10000	3	0.0773	-0.0150/0.241
hybrid pinball normalized*lr tripled	collapsed	60/10000	3	-0.0833	-0.25/0.00
hybrid pinball normalized*lr tripled	rate	60/10000	3	0.0498	-0.0198/0.142
hybrid pinball normalized*lr tripled	distance	60/10000	3	0.0314	0.0108/0.0472
hybrid pinball normalized*lr tripled*clipping disabled	collapsed	60/10000	3	0.167	0.00/0.5
hybrid pinball normalized*lr tripled*clipping disabled	rate	60/10000	3	0.00555	-0.187/0.134
hybrid pinball normalized*lr tripled*clipping disabled	distance	60/10000	3	0.0451	-0.00740/0.130

Table S6a. Weighting attribution: all paired contrasts

Source: `real_weighting_attribution.csv`. Rows: 12. The full source is in `validation_data`.

Group / comparison	Fit / test engines	Before rate	After rate	Reduction, pp [95% CI]	Resolved / introduced
FD001 lstm weighted raw minus fixed weight correction	60/100	20.4%	18.0%	2.4 [-4.4, 9.8]	11.0%/8.6%
FD001 lstm weighted raw minus unweighted retrain	60/100	20.4%	18.8%	1.6 [-5.2, 9.0]	12.2%/10.6%
FD001 lstm corrected weighted minus unweighted retrain	60/100	18.0%	18.8%	-0.8 [-3.8, 1.8]	3.6%/4.4%
FD001 hybrid weighted raw minus fixed weight correction	60/100	13.6%	12.2%	1.4 [-2.8, 6.0]	8.6%/7.2%
FD001 hybrid weighted raw minus unweighted retrain	60/100	13.6%	21.6%	-8.0 [-14.2, -2.4]	6.2%/14.2%
FD001 hybrid corrected weighted minus unweighted retrain	60/100	12.2%	21.6%	-9.4 [-13.4, -5.6]	4.6%/14.0%
FD002 lstm weighted raw minus fixed weight correction	156/259	9.6%	3.5%	6.1 [3.5, 8.9]	8.6%/2.5%
FD002 hybrid weighted raw minus fixed weight correction	156/259	8.9%	6.6%	2.2 [-0.2, 4.8]	6.9%/4.6%
FD003 lstm weighted raw minus fixed weight correction	60/100	12.8%	7.0%	5.8 [1.0, 10.4]	10.2%/4.4%
FD003 hybrid weighted raw minus fixed weight correction	60/100	11.8%	8.6%	3.2 [-0.2, 7.0]	8.4%/5.2%
FD004 lstm weighted raw minus fixed weight correction	149/248	16.6%	5.9%	10.7 [7.6, 14.0]	13.6%/2.9%
FD004 hybrid weighted raw minus fixed weight correction	149/248	12.7%	13.2%	-0.6 [-4.2, 2.7]	11.0%/11.6%

Table S6b. Complete compatibility and Brier by probability method

Source: `tableS21_cross_head.md`.

Group/method	Test engines	Incompatible	Mean distance	Brier
FD001 LSTM Raw	100	20.4 ± 4.6%	0.0210 ± 0.0061	0.03561 ± 0.00429
FD001 LSTM TrainOff	100	18.0 ± 2.4%	0.0228 ± 0.0042	0.05276 ± 0.00569
FD001 LSTM Temp	100	31.4 ± 2.3%	0.0369 ± 0.0108	0.04191 ± 0.00368
FD001 LSTM TrainOff+Temp	100	19.2 ± 2.7%	0.0260 ± 0.0089	0.05722 ± 0.00906
FD001 Hybrid Raw	100	13.6 ± 3.9%	0.0138 ± 0.0075	0.03477 ± 0.00728
FD001 Hybrid TrainOff	100	12.2 ± 4.7%	0.0156 ± 0.0065	0.04555 ± 0.00874
FD001 Hybrid Temp	100	18.6 ± 5.0%	0.0197 ± 0.0097	0.03656 ± 0.00793

Group/method	Test engines	Incompatible	Mean distance	Brier
FD001 Hybrid TrainOff+Temp	100	13.2 ± 3.9%	0.0176 ± 0.0066	0.05030 ± 0.01049
FD002 LSTM Raw	259	9.6 ± 1.4%	0.0067 ± 0.0021	0.02510 ± 0.00343
FD002 LSTM TrainOff	259	3.5 ± 1.2%	0.0008 ± 0.0003	0.01361 ± 0.00301
FD002 LSTM Temp	259	11.2 ± 0.9%	0.0141 ± 0.0057	0.03357 ± 0.00673
FD002 LSTM TrainOff+Temp	259	11.4 ± 0.2%	0.0079 ± 0.0010	0.01507 ± 0.00702
FD002 Hybrid Raw	259	8.9 ± 2.7%	0.0072 ± 0.0040	0.03333 ± 0.01419
FD002 Hybrid TrainOff	259	6.6 ± 2.6%	0.0060 ± 0.0036	0.02090 ± 0.00967
FD002 Hybrid Temp	259	13.4 ± 6.9%	0.0129 ± 0.0075	0.03598 ± 0.01190
FD002 Hybrid TrainOff+Temp	259	12.8 ± 1.0%	0.0131 ± 0.0045	0.02314 ± 0.00930
FD003 LSTM Raw	100	12.8 ± 4.1%	0.0150 ± 0.0081	0.02666 ± 0.00860
FD003 LSTM TrainOff	100	7.0 ± 2.7%	0.0024 ± 0.0016	0.02669 ± 0.00261
FD003 LSTM Temp	100	25.0 ± 5.8%	0.0263 ± 0.0106	0.03403 ± 0.00747
FD003 LSTM TrainOff+Temp	100	11.0 ± 3.8%	0.0053 ± 0.0033	0.02708 ± 0.00329
FD003 Hybrid Raw	100	11.8 ± 3.3%	0.0149 ± 0.0113	0.02915 ± 0.01492
FD003 Hybrid TrainOff	100	8.6 ± 4.3%	0.0081 ± 0.0025	0.02207 ± 0.00476
FD003 Hybrid Temp	100	36.2 ± 12.4%	0.0398 ± 0.0160	0.03750 ± 0.01141
FD003 Hybrid TrainOff+Temp	100	23.8 ± 10.2%	0.0248 ± 0.0095	0.03268 ± 0.01069
FD004 LSTM Raw	248	16.6 ± 2.1%	0.0137 ± 0.0029	0.05290 ± 0.00297
FD004 LSTM TrainOff	248	5.9 ± 1.7%	0.0013 ± 0.0005	0.03499 ± 0.00236
FD004 LSTM Temp	248	16.7 ± 2.8%	0.0186 ± 0.0067	0.05904 ± 0.00580
FD004 LSTM TrainOff+Temp	248	19.6 ± 2.3%	0.0124 ± 0.0028	0.04141 ± 0.00720
FD004 Hybrid Raw	248	12.7 ± 2.5%	0.0086 ± 0.0031	0.04484 ± 0.00384
FD004 Hybrid TrainOff	248	13.2 ± 3.6%	0.0102 ± 0.0053	0.03787 ± 0.00415
FD004 Hybrid Temp	248	15.3 ± 6.6%	0.0126 ± 0.0072	0.05017 ± 0.00972
FD004 Hybrid TrainOff+Temp	248	19.8 ± 3.4%	0.0225 ± 0.0069	0.04513 ± 0.00622

Table S7a. All real associations, including incremental probability adjustment

Source: `real_associations.csv`. Rows: 152. The full source is in `validation_data`.

Group / metric	Fit / test engines	Incomp / comp engines	Effect [95% CI]	Undefined draws
FD001 lstm median squared error mean incompatible	60/100	35/93	175 [47.9, 407]	0
FD001 lstm median squared error mean compatible	60/100	35/93	228 [170, 294]	0
FD001 lstm median squared error unadjusted difference	60/100	35/93	-53.0 [-200, 179]	0
FD001 lstm Brier mean incompatible	60/100	35/93	0.10110 [0.05167, 0.15650]	0
FD001 lstm Brier mean compatible	60/100	35/93	0.01882 [0.00422, 0.04074]	0
FD001 lstm Brier unadjusted difference	60/100	35/93	0.0823 [0.0380, 0.133]	0
FD001 lstm median squared error adjusted flag	60/100	35/93	69.7 [-167, 405]	0
FD001 lstm Brier adjusted flag	60/100	35/93	0.0426 [-0.0369, 0.124]	0

Group / metric	Fit / test engines	Incomp / comp engines	Effect [95% CI]	Undefined draws
FD001 lstm median squared error adjusted distance per 0.1	60/100	35/93	79.3 [-112, 326]	0
FD001 lstm Brier adjusted distance per 0.1	60/100	35/93	0.0433 [-0.0228, 0.101]	0
FD001 lstm median squared error RUL bin standardized difference	60/100	35/93	712 [657, 768]	1219
FD001 lstm Brier RUL bin standardized difference	60/100	35/93	0.175 [0.133, 0.207]	1219
FD001 lstm near threshold rate 20 to 40	60/100	35/93	50.7% [33.3%, 68.0%]	0
FD001 lstm near threshold distance 20 to 40	60/100	35/93	0.0731 [0.0417, 0.104]	0
FD001 lstm near threshold rate elsewhere	60/100	35/93	15.1% [9.3%, 21.5%]	0
FD001 lstm near threshold distance elsewhere	60/100	35/93	0.0118 [0.00556, 0.0193]	0
FD001 lstm near threshold rate difference	60/100	35/93	35.6 [18.0, 54.0]	0
FD001 lstm Brier distance beyond raw p per 0.1	60/100	35/93	-0.000773 [-0.0394, 0.0398]	0
FD001 lstm Brier incremental in sample R squared	60/100	35/93	0.00000656 [0.00000612, 0.0449]	0
FD001 hybrid median squared error mean incompatible	60/100	25/98	104 [40.9, 193]	0
FD001 hybrid median squared error mean compatible	60/100	25/98	283 [210, 367]	0
FD001 hybrid median squared error unadjusted difference	60/100	25/98	-179 [-267, -96.0]	0
FD001 hybrid Brier mean incompatible	60/100	25/98	0.09590 [0.04213, 0.17398]	0
FD001 hybrid Brier mean compatible	60/100	25/98	0.02515 [0.00758, 0.04921]	0
FD001 hybrid Brier unadjusted difference	60/100	25/98	0.0708 [0.0266, 0.141]	0
FD001 hybrid median squared error adjusted flag	60/100	25/98	-250 [-415, -95.3]	0
FD001 hybrid Brier adjusted flag	60/100	25/98	-0.0143 [-0.0887, 0.0550]	0
FD001 hybrid median squared error adjusted distance per 0.1	60/100	25/98	-99.2 [-182, -33.0]	0
FD001 hybrid Brier adjusted distance per 0.1	60/100	25/98	0.0203 [-0.0276, 0.0710]	0
FD001 hybrid median squared error RUL bin standardized difference	60/100	25/98	416 [310, 484]	1266
FD001 hybrid Brier RUL bin standardized difference	60/100	25/98	0.117 [0.0703, 0.151]	1266
FD001 hybrid near threshold rate 20 to 40	60/100	25/98	53.3% [36.7%, 68.6%]	0

Group / metric	Fit / test engines	Incomp / comp engines	Effect [95% CI]	Undefined draws
FD001 hybrid near threshold distance 20 to 40	60/100	25/98	0.0563 [0.0333, 0.0832]	0
FD001 hybrid near threshold rate elsewhere	60/100	25/98	6.6% [3.0%, 10.9%]	0
FD001 hybrid near threshold distance elsewhere	60/100	25/98	0.00629 [0.00222, 0.0114]	0
FD001 hybrid near threshold rate difference	60/100	25/98	46.7 [29.7, 62.9]	0
FD001 hybrid Brier distance beyond raw p per 0.1	60/100	25/98	-0.0215 [-0.0772, 0.0331]	0
FD001 hybrid Brier incremental in sample R squared	60/100	25/98	0.00514 [0.0000230, 0.0498]	0
FD002 lstm median squared error mean incompatible	156/259	46/255	51.7 [34.5, 71.4]	0
FD002 lstm median squared error mean compatible	156/259	46/255	222 [171, 280]	0
FD002 lstm median squared error unadjusted difference	156/259	46/255	-170 [-232, -118]	0
FD002 lstm Brier mean incompatible	156/259	46/255	0.15612 [0.07903, 0.23981]	0
FD002 lstm Brier mean compatible	156/259	46/255	0.01122 [0.00468, 0.02001]	0
FD002 lstm Brier unadjusted difference	156/259	46/255	0.145 [0.0724, 0.226]	0
FD002 lstm median squared error adjusted flag	156/259	46/255	-138 [-240, -51.0]	0
FD002 lstm Brier adjusted flag	156/259	46/255	0.0908 [0.0332, 0.155]	0
FD002 lstm median squared error adjusted distance per 0.1	156/259	46/255	-83.3 [-162, -31.7]	0
FD002 lstm Brier adjusted distance per 0.1	156/259	46/255	0.123 [0.0631, 0.182]	0
FD002 lstm median squared error RUL bin standardized difference	156/259	46/255	-14.0 [-87.5, 48.4]	735
FD002 lstm Brier RUL bin standardized difference	156/259	46/255	0.0393 [0.00617, 0.0740]	735
FD002 lstm near threshold rate 20 to 40	156/259	46/255	54.5% [38.9%, 69.6%]	0
FD002 lstm near threshold distance 20 to 40	156/259	46/255	0.0514 [0.0320, 0.0729]	0
FD002 lstm near threshold rate elsewhere	156/259	46/255	5.4% [3.4%, 7.7%]	0
FD002 lstm near threshold distance elsewhere	156/259	46/255	0.00256 [0.00138, 0.00402]	0
FD002 lstm near threshold rate difference	156/259	46/255	49.1 [33.3, 64.2]	0
FD002 lstm Brier distance beyond raw p per 0.1	156/259	46/255	0.0199 [-0.0233, 0.0724]	0

Group / metric	Fit / test engines	Incomp / comp engines	Effect [95% CI]	Undefined draws
FD002 lstm Brier incremental in sample R squared	156/259	46/255	0.00190 [0.00000397, 0.0257]	0
FD002 hybrid median squared error mean incompatible	156/259	49/256	118 [67.4, 189]	0
FD002 hybrid median squared error mean compatible	156/259	49/256	252 [198, 312]	0
FD002 hybrid median squared error unadjusted difference	156/259	49/256	-134 [-208, -62.2]	0
FD002 hybrid Brier mean incompatible	156/259	49/256	0.19796 [0.12007, 0.26964]	0
FD002 hybrid Brier mean compatible	156/259	49/256	0.01729 [0.00845, 0.02851]	0
FD002 hybrid Brier unadjusted difference	156/259	49/256	0.181 [0.106, 0.252]	0
FD002 hybrid median squared error adjusted flag	156/259	49/256	-149 [-311, -4.99]	0
FD002 hybrid Brier adjusted flag	156/259	49/256	0.103 [0.0365, 0.171]	0
FD002 hybrid median squared error adjusted distance per 0.1	156/259	49/256	-97.0 [-173, -43.6]	0
FD002 hybrid Brier adjusted distance per 0.1	156/259	49/256	0.137 [0.0907, 0.171]	0
FD002 hybrid median squared error RUL bin standardized difference	156/259	49/256	947 [881, 994]	1221
FD002 hybrid Brier RUL bin standardized difference	156/259	49/256	0.0296 [0.0114, 0.0487]	1221
FD002 hybrid near threshold rate 20 to 40	156/259	49/256	41.8% [30.0%, 53.0%]	0
FD002 hybrid near threshold distance 20 to 40	156/259	49/256	0.0432 [0.0225, 0.0645]	0
FD002 hybrid near threshold rate elsewhere	156/259	49/256	5.8% [3.7%, 8.1%]	0
FD002 hybrid near threshold distance elsewhere	156/259	49/256	0.00386 [0.00173, 0.00652]	0
FD002 hybrid near threshold rate difference	156/259	49/256	36.0 [23.7, 47.6]	0
FD002 hybrid Brier distance beyond raw p per 0.1	156/259	49/256	0.0596 [0.0104, 0.106]	0
FD002 hybrid Brier incremental in sample R squared	156/259	49/256	0.0202 [0.000443, 0.0936]	0
FD003 lstm median squared error mean incompatible	60/100	26/97	88.9 [49.4, 143]	0
FD003 lstm median squared error mean compatible	60/100	26/97	358 [231, 500]	0
FD003 lstm median squared error unadjusted difference	60/100	26/97	-269 [-411, -137]	0

Group / metric	Fit / test engines	Incomp / comp engines	Effect [95% CI]	Undefined draws
FD003 lstm Brier mean incompatible	60/100	26/97	0.13123 [0.03956, 0.25404]	0
FD003 lstm Brier mean compatible	60/100	26/97	0.01131 [0.00355, 0.02339]	0
FD003 lstm Brier unadjusted difference	60/100	26/97	0.120 [0.0340, 0.237]	0
FD003 lstm median squared error adjusted flag	60/100	26/97	-320 [-595, -71.0]	0
FD003 lstm Brier adjusted flag	60/100	26/97	0.0822 [0.0155, 0.179]	0
FD003 lstm median squared error adjusted distance per 0.1	60/100	26/97	-110 [-226, -22.3]	0
FD003 lstm Brier adjusted distance per 0.1	60/100	26/97	0.0471 [0.00669, 0.111]	0
FD003 lstm median squared error RUL bin standardized difference	60/100	26/97	-92.4 [-261, 56.8]	934
FD003 lstm Brier RUL bin standardized difference	60/100	26/97	0.0427 [-0.000513, 0.106]	934
FD003 lstm near threshold rate 20 to 40	60/100	26/97	62.0% [48.9%, 76.7%]	0
FD003 lstm near threshold distance 20 to 40	60/100	26/97	0.0871 [0.0450, 0.139]	0
FD003 lstm near threshold rate elsewhere	60/100	26/97	7.3% [3.7%, 11.4%]	0
FD003 lstm near threshold distance elsewhere	60/100	26/97	0.00696 [0.00262, 0.0127]	0
FD003 lstm near threshold rate difference	60/100	26/97	54.7 [41.0, 69.2]	0
FD003 lstm Brier distance beyond raw p per 0.1	60/100	26/97	-0.00252 [-0.0408, 0.0500]	0
FD003 lstm Brier incremental in sample R squared	60/100	26/97	0.000105 [0.0000151, 0.0570]	0
FD003 hybrid median squared error mean incompatible	60/100	28/99	111 [50.8, 197]	0
FD003 hybrid median squared error mean compatible	60/100	28/99	251 [183, 329]	0
FD003 hybrid median squared error unadjusted difference	60/100	28/99	-139 [-238, -35.8]	0
FD003 hybrid Brier mean incompatible	60/100	28/99	0.17405 [0.07676, 0.27455]	0
FD003 hybrid Brier mean compatible	60/100	28/99	0.00976 [0.00398, 0.01753]	0
FD003 hybrid Brier unadjusted difference	60/100	28/99	0.164 [0.0708, 0.262]	0
FD003 hybrid median squared error adjusted flag	60/100	28/99	-99.9 [-230, 29.0]	0
FD003 hybrid Brier adjusted flag	60/100	28/99	0.129 [0.0524, 0.209]	0

Group / metric	Fit / test engines	Incomp / comp engines	Effect [95% CI]	Undefined draws
FD003 hybrid median squared error adjusted distance per 0.1	60/100	28/99	-17.0 [-71.4, 37.2]	0
FD003 hybrid Brier adjusted distance per 0.1	60/100	28/99	0.0743 [0.0366, 0.119]	0
FD003 hybrid median squared error RUL bin standardized difference	60/100	28/99	-5.95 [-90.0, 71.9]	342
FD003 hybrid Brier RUL bin standardized difference	60/100	28/99	0.0904 [0.0355, 0.132]	342
FD003 hybrid near threshold rate 20 to 40	60/100	28/99	52.0% [35.6%, 70.0%]	0
FD003 hybrid near threshold distance 20 to 40	60/100	28/99	0.0782 [0.0402, 0.114]	0
FD003 hybrid near threshold rate elsewhere	60/100	28/99	7.3% [4.2%, 11.1%]	0
FD003 hybrid near threshold distance elsewhere	60/100	28/99	0.00791 [0.00336, 0.0133]	0
FD003 hybrid near threshold rate difference	60/100	28/99	44.7 [28.2, 62.4]	0
FD003 hybrid Brier distance beyond raw p per 0.1	60/100	28/99	0.0150 [-0.0179, 0.0504]	0
FD003 hybrid Brier incremental in sample R squared	60/100	28/99	0.00385 [0.00000821, 0.0456]	0
FD004 lstm median squared error mean incompatible	149/248	69/236	176 [82.9, 330]	0
FD004 lstm median squared error mean compatible	149/248	69/236	295 [229, 366]	0
FD004 lstm median squared error unadjusted difference	149/248	69/236	-119 [-234, 29.2]	0
FD004 lstm Brier mean incompatible	149/248	69/236	0.21846 [0.13797, 0.30776]	0
FD004 lstm Brier mean compatible	149/248	69/236	0.01992 [0.01072, 0.03108]	0
FD004 lstm Brier unadjusted difference	149/248	69/236	0.199 [0.121, 0.288]	0
FD004 lstm median squared error adjusted flag	149/248	69/236	-36.3 [-276, 258]	0
FD004 lstm Brier adjusted flag	149/248	69/236	0.124 [0.0554, 0.202]	0
FD004 lstm median squared error adjusted distance per 0.1	149/248	69/236	37.0 [-166, 301]	0
FD004 lstm Brier adjusted distance per 0.1	149/248	69/236	0.107 [0.0572, 0.154]	0
FD004 lstm median squared error RUL bin standardized difference	149/248	69/236	545 [60.3, 853]	121
FD004 lstm Brier RUL bin standardized difference	149/248	69/236	0.143 [0.0379, 0.208]	121
FD004 lstm near threshold rate 20 to 40	149/248	69/236	62.8% [51.1%, 74.5%]	0

Group / metric	Fit / test engines	Incomp / comp engines	Effect [95% CI]	Undefined draws
FD004 lstm near threshold distance 20 to 40	149/248	69/236	0.0698 [0.0488, 0.0925]	0
FD004 lstm near threshold rate elsewhere	149/248	69/236	10.5% [7.4%, 13.9%]	0
FD004 lstm near threshold distance elsewhere	149/248	69/236	0.00630 [0.00409, 0.00902]	0
FD004 lstm near threshold rate difference	149/248	69/236	52.3 [40.3, 64.4]	0
FD004 lstm Brier distance beyond raw p per 0.1	149/248	69/236	0.0385 [-0.00714, 0.0892]	0
FD004 lstm Brier incremental in sample R squared	149/248	69/236	0.00684 [0.0000225, 0.0351]	0
FD004 hybrid median squared error mean incompatible	149/248	72/245	215 [75.9, 475]	0
FD004 hybrid median squared error mean compatible	149/248	72/245	282 [224, 346]	0
FD004 hybrid median squared error unadjusted difference	149/248	72/245	-67.2 [-230, 191]	0
FD004 hybrid Brier mean incompatible	149/248	72/245	0.17295 [0.09377, 0.26135]	0
FD004 hybrid Brier mean compatible	149/248	72/245	0.02627 [0.01450, 0.03968]	0
FD004 hybrid Brier unadjusted difference	149/248	72/245	0.147 [0.0696, 0.233]	0
FD004 hybrid median squared error adjusted flag	149/248	72/245	90.2 [-147, 429]	0
FD004 hybrid Brier adjusted flag	149/248	72/245	0.103 [0.0247, 0.187]	0
FD004 hybrid median squared error adjusted distance per 0.1	149/248	72/245	234 [-140, 670]	0
FD004 hybrid Brier adjusted distance per 0.1	149/248	72/245	0.128 [0.0264, 0.186]	0
FD004 hybrid median squared error RUL bin standardized difference	149/248	72/245	622 [105, 882]	756
FD004 hybrid Brier RUL bin standardized difference	149/248	72/245	0.160 [0.0200, 0.229]	756
FD004 hybrid near threshold rate 20 to 40	149/248	72/245	31.7% [20.8%, 43.4%]	0
FD004 hybrid near threshold distance 20 to 40	149/248	72/245	0.0233 [0.0106, 0.0430]	0
FD004 hybrid near threshold rate elsewhere	149/248	72/245	10.1% [7.5%, 13.2%]	0
FD004 hybrid near threshold distance elsewhere	149/248	72/245	0.00662 [0.00323, 0.0112]	0
FD004 hybrid near threshold rate difference	149/248	72/245	21.6 [10.4, 33.4]	0
FD004 hybrid Brier distance beyond raw p per 0.1	149/248	72/245	0.0399 [-0.0328, 0.121]	0

Group / metric	Fit / test engines	Incomp / comp engines	Effect [95% CI]	Undefined draws
FD004 hybrid Brier incremental in sample R squared	149/248	72/245	0.00742 [0.0000179, 0.0706]	0

Table S7b. All pre-specified real RUL-bin contrasts

Source: `real_RUL_bins.csv`. Rows: 48. The full source is in `validation_data`.

Group / RUL bin	Fit / test / bin engines	Incomp / comp engines	Median error difference [95% CI]	Brier difference [95% CI]
FD001 lstm [0,20)	60/100/13	12/11	49.7 [-9.68, 150]	0.0187 [0.00150, 0.0507]
FD001 lstm [20,40]	60/100/15	13/12	35.3 [-47.9, 115]	-0.0313 [-0.195, 0.0754]
FD001 lstm (40,60]	60/100/11	8/9	-191 [-305, -87.7]	0.0563 [-0.0290, 0.158]
FD001 lstm (60,80]	60/100/6	1/6	-358 [-732, 194]	0.0494 [0.0493, 0.0494]
FD001 lstm (80,100]	60/100/22	1/22	2.31e+3 [2.22e+3, 2.36e+3]	0.503 [0.498, 0.503]
FD001 lstm (100,125]	60/100/33	0/33	NA [NA, NA]	NA [NA, NA]
FD001 hybrid [0,20)	60/100/13	3/12	48.8 [-15.7, 75.2]	0.0273 [-0.00449, 0.0400]
FD001 hybrid [20,40]	60/100/15	13/14	-63.8 [-199, 32.0]	-0.155 [-0.346, -0.0148]
FD001 hybrid (40,60]	60/100/11	7/11	-393 [-640, -96.6]	0.0714 [0.0114, 0.152]
FD001 hybrid (60,80]	60/100/6	1/6	-637 [-1.12e+3, 59.4]	0.0422 [0.0308, 0.0432]
FD001 hybrid (80,100]	60/100/22	1/22	1.65e+3 [1.43e+3, 1.71e+3]	0.400 [0.381, 0.401]
FD001 hybrid (100,125]	60/100/33	0/33	NA [NA, NA]	NA [NA, NA]
FD002 lstm [0,20)	156/259/53	12/53	31.0 [15.9, 46.3]	0.00363 [0.000236, 0.0114]
FD002 lstm [20,40]	156/259/22	18/18	-110 [-298, 11.7]	0.0871 [-0.0606, 0.229]
FD002 lstm (40,60]	156/259/27	15/27	-159 [-351, -9.76]	0.101 [-0.00281, 0.231]
FD002 lstm (60,80]	156/259/28	1/28	115 [-138, 330]	0.00978 [0.00964, 0.00986]
FD002 lstm (80,100]	156/259/37	0/37	NA [NA, NA]	NA [NA, NA]
FD002 lstm (100,125]	156/259/92	0/92	NA [NA, NA]	NA [NA, NA]
FD002 hybrid [0,20)	156/259/53	9/53	7.41 [-5.50, 21.7]	0.000790 [-0.000792, 0.00296]
FD002 hybrid [20,40]	156/259/22	18/21	-85.4 [-242, 11.1]	0.0610 [-0.0929, 0.213]
FD002 hybrid (40,60]	156/259/27	16/25	-239 [-514, -25.1]	0.165 [0.0460, 0.301]
FD002 hybrid (60,80]	156/259/28	4/28	-72.9 [-464, 243]	0.00862 [0.000578, 0.0425]
FD002 hybrid (80,100]	156/259/37	1/37	764 [666, 858]	0.0378 [0.0378, 0.0378]
FD002 hybrid (100,125]	156/259/92	1/92	2.47e+3 [2.35e+3, 2.53e+3]	0.00220 [0.000984, 0.00249]
FD003 lstm [0,20)	60/100/13	5/11	56.4 [0.309, 145]	0.0139 [-0.00381, 0.0967]
FD003 lstm [20,40]	60/100/10	10/9	-0.354 [-26.3, 23.4]	0.0131 [-0.132, 0.158]
FD003 lstm (40,60]	60/100/18	8/18	-673 [-1.17e+3, -213]	0.131 [0.00757, 0.346]
FD003 lstm (60,80]	60/100/12	2/12	76.5 [-486, 437]	0.0162 [0.0137, 0.0174]
FD003 lstm (80,100]	60/100/15	1/15	279 [101, 416]	0.00262 [0.00260, 0.00263]
FD003 lstm (100,125]	60/100/32	0/32	NA [NA, NA]	NA [NA, NA]
FD003 hybrid [0,20)	60/100/13	2/13	46.3 [21.4, 90.1]	0.000139 [-0.00104, 0.000557]
FD003 hybrid [20,40]	60/100/10	10/9	11.5 [-2.25, 26.3]	0.133 [-0.0306, 0.248]
FD003 hybrid (40,60]	60/100/18	13/18	-232 [-405, -68.5]	0.101 [0.0218, 0.172]
FD003 hybrid (60,80]	60/100/12	3/12	262 [-10.3, 482]	0.137 [0.0110, 0.243]
FD003 hybrid (80,100]	60/100/15	0/15	NA [NA, NA]	NA [NA, NA]
FD003 hybrid (100,125]	60/100/32	0/32	NA [NA, NA]	NA [NA, NA]
FD004 lstm [0,20)	149/248/40	20/35	15.7 [3.15, 32.4]	0.000152 [-0.0000508, 0.000352]

Group / RUL bin	Fit / test / bin engines	Incomp / comp engines	Median error difference [95% CI]	Brier difference [95% CI]
FD004 lstm [20,40]	149/248/29	26/22	38.0 [-31.6, 109]	0.230 [0.0784, 0.380]
FD004 lstm (40,60]	149/248/22	16/22	-483 [-907, -90.0]	0.0863 [-0.00970, 0.194]
FD004 lstm (60,80]	149/248/22	4/22	-153 [-466, 150]	0.0161 [0.00869, 0.0241]
FD004 lstm (80,100]	149/248/35	3/35	2.65e+3 [763, 3.90e+3]	0.352 [0.00292, 0.587]
FD004 lstm (100,125]	149/248/100	0/100	NA [NA, NA]	NA [NA, NA]
FD004 hybrid [0,20)	149/248/40	31/39	4.27 [-11.0, 17.8]	-0.000760 [-0.00202, 0.000178]
FD004 hybrid [20,40]	149/248/29	20/28	86.0 [9.77, 202]	0.149 [-0.0322, 0.329]
FD004 hybrid (40,60]	149/248/22	17/21	-487 [-941, -88.3]	0.150 [0.00367, 0.321]
FD004 hybrid (60,80]	149/248/22	1/22	-178 [-386, 27.7]	0.00610 [0.00593, 0.00619]
FD004 hybrid (80,100]	149/248/35	3/35	2.97e+3 [928, 3.98e+3]	0.457 [0.00337, 0.684]
FD004 hybrid (100,125]	149/248/100	0/100	NA [NA, NA]	NA [NA, NA]

Table S8. All primary family-level oracle association coefficients

Source: `synthetic_family_confirmation.csv`. Rows: 8. The full source is in `validation_data`.

Family / outcome / predictor	Fit engines	Test engines / subset	Coefficient [95% CI]	Pass
temporal oracle probability squared deviation flag	60/156/60/149	10000	0.00501 [0.00492, 0.00511]	True
temporal oracle probability squared deviation distance per 0.1	60/156/60/149	10000	0.00141 [0.00138, 0.00145]	True
temporal oracle quantile MAE flag	60/156/60/149	10000	1.21 [1.19, 1.24]	True
temporal oracle quantile MAE distance per 0.1	60/156/60/149	10000	0.428 [0.420, 0.437]	True
covariate oracle probability squared deviation flag	60/156/60/149	10000	-0.00124 [-0.00151, -0.000952]	False
covariate oracle probability squared deviation distance per 0.1	60/156/60/149	10000	0.000923 [0.000811, 0.00104]	True
covariate oracle quantile MAE flag	60/156/60/149	10000	-1.89 [-1.91, -1.86]	False
covariate oracle quantile MAE distance per 0.1	60/156/60/149	10000	-0.655 [-0.670, -0.640]	False

All setting-specific oracle coefficients, group means and sensitivity results are retained in the full-precision companion `synthetic_associations.csv`. Table S8 contains every primary confirmation component.

Table S9a. Real Brier flag association after probability, band and RUL adjustment

Source: `real_Brier_adjusted.csv`. Rows: 8. The full source is in `validation_data`.

Group	Fit / test engines	Band / elsewhere engines	Incomp / comp engines	Coefficient [95% CI]	Pass
FD001 lstm	60/100	15/85	35/93	0.0242 [-0.0271, 0.0765]	False
FD001 hybrid	60/100	15/85	25/98	-0.110 [-0.206, -0.0112]	False
FD002 lstm	156/259	22/237	46/255	-0.00565 [-0.0482, 0.0400]	False

Group	Fit / test engines	Band / elsewhere engines	Incomp / comp engines	Coefficient [95% CI]	Pass
FD002 hybrid	156/259	22/237	49/256	0.0128 [-0.0417, 0.0661]	False
FD003 lstm	60/100	10/90	26/97	-0.0105 [-0.0733, 0.0527]	False
FD003 hybrid	60/100	10/90	28/99	0.0481 [0.00764, 0.0878]	True
FD004 lstm	149/248	29/219	69/236	0.0145 [-0.0519, 0.0796]	False
FD004 hybrid	149/248	29/219	72/245	0.0541 [-0.0116, 0.120]	False

147 ## Table S9b. Real near-threshold and elsewhere rates

148 Source: `near_threshold_table.csv`. Rows: 8. The full source is in `validation_data`.

Group	Fit / test engines	Band / elsewhere engines	Band rate [95% CI]	Elsewhere rate [95% CI]
FD001 lstm	60/100	15/85	50.7% [33.3%, 68.0%]	15.1% [9.3%, 21.5%]
FD001 hybrid	60/100	15/85	53.3% [36.7%, 68.6%]	6.6% [3.0%, 10.9%]
FD002 lstm	156/259	22/237	54.5% [38.9%, 69.6%]	5.4% [3.4%, 7.7%]
FD002 hybrid	156/259	22/237	41.8% [30.0%, 53.0%]	5.8% [3.7%, 8.1%]
FD003 lstm	60/100	10/90	62.0% [48.9%, 76.7%]	7.3% [3.7%, 11.4%]
FD003 hybrid	60/100	10/90	52.0% [35.6%, 70.0%]	7.3% [4.2%, 11.1%]
FD004 lstm	149/248	29/219	62.8% [51.1%, 74.5%]	10.5% [7.4%, 13.9%]
FD004 hybrid	149/248	29/219	31.7% [20.8%, 43.4%]	10.1% [7.5%, 13.2%]

149 ## Table S10. All synthetic near-threshold rates, both endpoint policies and fleet sizes

150 Source: `synthetic_band_rates_wide.csv`. Rows: 144. The full source is in `validation_data`.

Group / variant / policy	Fit / test engines	Band / elsewhere engines	Fits	Band rate [95% CI]	Elsewhere rate [95% CI]	Difference, pp [95% CI]
FD001 lstm weighted raw final	60/100	24/76	30	68.5% [67.8%, 69.1%]	67.5% [67.1%, 67.8%]	1.0 [0.2, 1.8]
FD001 lstm weighted raw final	60/10000	2062/7938	30	67.8% [67.7%, 67.8%]	67.5% [67.4%, 67.5%]	0.3 [0.2, 0.4]
FD001 lstm weighted corrected final	60/100	24/76	30	54.9% [53.6%, 56.2%]	53.6% [53.0%, 54.1%]	1.3 [-0.1, 2.7]
FD001 lstm weighted corrected final	60/10000	2062/7938	30	53.6% [53.5%, 53.8%]	53.5% [53.4%, 53.6%]	0.1 [-0.0, 0.3]
FD001 lstm unweighted final	60/100	24/76	30	55.1% [53.8%, 56.5%]	53.7% [53.2%, 54.3%]	1.4 [-0.0, 3.0]
FD001 lstm unweighted final	60/10000	2062/7938	30	54.0% [53.9%, 54.1%]	53.8% [53.7%, 53.8%]	0.2 [0.1, 0.4]
FD001 lstm weighted raw uniform	60/100	22/78	30	67.7% [67.1%, 68.4%]	68.1% [67.6%, 68.6%]	-0.4 [-1.2, 0.4]
FD001 lstm weighted raw uniform	60/10000	1734/8266	30	68.3% [68.2%, 68.4%]	68.4% [68.3%, 68.4%]	-0.0 [-0.2, 0.1]
FD001 lstm weighted corrected uniform	60/100	22/78	30	56.5% [55.1%, 58.1%]	56.5% [55.7%, 57.3%]	0.0 [-1.7, 1.8]

Group / variant / policy	Fit / test engines	Band / elsewhere engines	Fits	Band rate [95% CI]	Elsewhere rate [95% CI]	Difference, pp [95% CI]
FD001 lstm weighted corrected uniform	60/10000	1734/8266	30	56.4% [56.2%, 56.5%]	56.6% [56.5%, 56.7%]	-0.2 [-0.4, -0.1]
FD001 lstm unweighted uniform	60/100	22/78	30	56.2% [54.9%, 57.7%]	56.4% [55.6%, 57.1%]	-0.2 [-1.7, 1.4]
FD001 lstm unweighted uniform	60/10000	1734/8266	30	56.4% [56.3%, 56.6%]	56.5% [56.5%, 56.6%]	-0.1 [-0.3, 0.1]
FD001 hybrid weighted raw final	60/100	24/76	30	58.5% [53.3%, 64.3%]	66.0% [62.3%, 69.6%]	-7.5 [-14.0, -0.8]
FD001 hybrid weighted raw final	60/10000	2062/7938	30	62.2% [61.5%, 62.9%]	66.6% [66.2%, 66.9%]	-4.3 [-5.1, -3.5]
FD001 hybrid weighted corrected final	60/100	24/76	30	93.9% [91.3%, 96.3%]	89.8% [88.4%, 91.4%]	4.1 [1.1, 6.9]
FD001 hybrid weighted corrected final	60/10000	2062/7938	30	91.5% [91.2%, 91.8%]	89.6% [89.5%, 89.8%]	1.9 [1.6, 2.2]
FD001 hybrid unweighted final	60/100	24/76	30	95.1% [92.2%, 97.8%]	90.8% [89.1%, 92.6%]	4.3 [1.0, 7.6]
FD001 hybrid unweighted final	60/10000	2062/7938	30	93.0% [92.7%, 93.4%]	90.8% [90.7%, 91.0%]	2.2 [1.8, 2.6]
FD001 hybrid weighted raw uniform	60/100	22/78	30	68.6% [61.8%, 74.9%]	67.1% [63.4%, 70.6%]	1.5 [-5.9, 9.2]
FD001 hybrid weighted raw uniform	60/10000	1734/8266	30	67.3% [66.6%, 68.0%]	66.8% [66.4%, 67.1%]	0.6 [-0.3, 1.4]
FD001 hybrid weighted corrected uniform	60/100	22/78	30	90.2% [87.1%, 93.5%]	90.2% [88.7%, 91.9%]	-0.1 [-3.6, 3.6]
FD001 hybrid weighted corrected uniform	60/10000	1734/8266	30	90.7% [90.4%, 91.1%]	90.8% [90.6%, 90.9%]	-0.0 [-0.4, 0.4]
FD001 hybrid unweighted uniform	60/100	22/78	30	90.3% [87.2%, 93.8%]	90.1% [88.6%, 91.7%]	0.2 [-3.4, 3.9]
FD001 hybrid unweighted uniform	60/10000	1734/8266	30	90.7% [90.3%, 91.0%]	90.6% [90.4%, 90.8%]	0.1 [-0.3, 0.5]
FD001 covariate weighted raw final	60/100	24/76	200	49.4% [37.9%, 59.6%]	33.7% [26.8%, 40.9%]	15.7 [2.5, 28.2]
FD001 covariate weighted raw final	60/10000	2062/7938	200	40.9% [39.6%, 42.3%]	32.4% [31.7%, 33.1%]	8.5 [7.0, 10.1]
FD001 covariate weighted corrected final	60/100	24/76	200	25.1% [18.9%, 30.6%]	16.4% [12.7%, 20.2%]	8.7 [1.7, 15.5]
FD001 covariate weighted corrected final	60/10000	2062/7938	200	20.3% [19.6%, 21.0%]	15.7% [15.3%, 16.1%]	4.6 [3.8, 5.4]
FD001 covariate unweighted final	60/100	24/76	200	27.4% [20.6%, 33.4%]	18.0% [14.1%, 22.2%]	9.4 [1.7, 16.7]
FD001 covariate unweighted final	60/10000	2062/7938	200	22.3% [21.6%, 23.1%]	17.3% [16.9%, 17.7%]	5.0 [4.1, 5.9]
FD001 covariate weighted raw uniform	60/100	22/78	200	31.5% [18.8%, 45.2%]	33.0% [26.3%, 40.0%]	-1.6 [-16.6, 13.3]

Group / variant / policy	Fit / test engines	Band / elsewhere engines	Fits	Band rate [95% CI]	Elsewhere rate [95% CI]	Difference, pp [95% CI]
FD001 covariate weighted raw uniform	60/10000	1734/8266	200	34.0% [32.5%, 35.5%]	34.7% [34.0%, 35.4%]	-0.8 [-2.4, 0.9]
FD001 covariate weighted corrected uniform	60/100	22/78	200	15.1% [8.4%, 22.3%]	15.7% [12.2%, 19.4%]	-0.6 [-8.4, 7.1]
FD001 covariate weighted corrected uniform	60/10000	1734/8266	200	16.4% [15.6%, 17.2%]	16.7% [16.3%, 17.0%]	-0.3 [-1.1, 0.6]
FD001 covariate unweighted uniform	60/100	22/78	200	16.3% [9.1%, 24.2%]	16.8% [13.0%, 20.7%]	-0.4 [-8.9, 7.9]
FD001 covariate unweighted uniform	60/10000	1734/8266	200	17.5% [16.7%, 18.4%]	17.7% [17.3%, 18.1%]	-0.2 [-1.1, 0.7]
FD002 lstm weighted raw final	156/259	53/206	30	34.5% [28.0%, 41.0%]	25.4% [21.9%, 28.8%]	9.1 [1.7, 16.7]
FD002 lstm weighted raw final	156/10000	2052/7948	30	32.3% [31.3%, 33.3%]	25.3% [24.8%, 25.8%]	7.0 [5.8, 8.1]
FD002 lstm weighted corrected final	156/259	53/206	30	22.8% [17.6%, 27.8%]	15.5% [12.8%, 18.0%]	7.3 [1.7, 13.0]
FD002 lstm weighted corrected final	156/10000	2052/7948	30	20.4% [19.6%, 21.2%]	15.2% [14.9%, 15.6%]	5.1 [4.2, 6.1]
FD002 lstm unweighted final	156/259	53/206	30	23.8% [18.5%, 29.0%]	16.0% [13.3%, 18.7%]	7.8 [2.0, 13.9]
FD002 lstm unweighted final	156/10000	2052/7948	30	20.8% [20.0%, 21.6%]	15.5% [15.1%, 15.9%]	5.3 [4.4, 6.2]
FD002 lstm weighted raw uniform	156/259	42/217	30	36.0% [27.3%, 44.7%]	33.9% [30.0%, 37.9%]	2.0 [-7.6, 11.7]
FD002 lstm weighted raw uniform	156/10000	1761/8239	30	35.0% [33.6%, 36.3%]	35.9% [35.2%, 36.5%]	-0.9 [-2.4, 0.6]
FD002 lstm weighted corrected uniform	156/259	42/217	30	23.1% [16.1%, 30.1%]	21.6% [18.5%, 24.7%]	1.5 [-6.1, 9.3]
FD002 lstm weighted corrected uniform	156/10000	1761/8239	30	22.4% [21.3%, 23.4%]	23.0% [22.5%, 23.5%]	-0.6 [-1.8, 0.6]
FD002 lstm unweighted uniform	156/259	42/217	30	23.5% [16.3%, 30.6%]	22.2% [19.0%, 25.5%]	1.2 [-6.6, 9.1]
FD002 lstm unweighted uniform	156/10000	1761/8239	30	22.9% [21.8%, 24.0%]	23.5% [23.0%, 24.0%]	-0.6 [-1.8, 0.6]
FD002 hybrid weighted raw final	156/259	53/206	30	40.8% [37.2%, 44.2%]	45.0% [43.4%, 46.7%]	-4.3 [-8.1, -0.6]
FD002 hybrid weighted raw final	156/10000	2052/7948	30	42.8% [42.3%, 43.3%]	45.6% [45.3%, 45.8%]	-2.8 [-3.4, -2.2]
FD002 hybrid weighted corrected final	156/259	53/206	30	71.9% [69.9%, 74.1%]	70.6% [69.7%, 71.7%]	1.2 [-1.1, 3.6]
FD002 hybrid weighted corrected final	156/10000	2052/7948	30	72.8% [72.5%, 73.1%]	71.3% [71.1%, 71.4%]	1.5 [1.2, 1.9]

Group / variant / policy	Fit / test engines	Band / elsewhere engines	Fits	Band rate [95% CI]	Elsewhere rate [95% CI]	Difference, pp [95% CI]
FD002 hybrid unweighted final	156/259	53/206	30	70.5% [68.4%, 72.7%]	68.9% [67.9%, 70.0%]	1.6 [-0.9, 4.0]
FD002 hybrid unweighted final	156/10000	2052/7948	30	71.2% [70.9%, 71.6%]	69.4% [69.2%, 69.5%]	1.9 [1.5, 2.3]
FD002 hybrid weighted raw uniform	156/259	42/217	30	38.5% [33.6%, 43.2%]	38.8% [36.8%, 41.0%]	-0.3 [-5.6, 4.7]
FD002 hybrid weighted raw uniform	156/10000	1761/8239	30	38.9% [38.1%, 39.6%]	38.2% [37.8%, 38.5%]	0.7 [-0.1, 1.5]
FD002 hybrid weighted corrected uniform	156/259	42/217	30	66.4% [65.3%, 67.7%]	65.6% [65.3%, 66.0%]	0.8 [-0.4, 2.1]
FD002 hybrid weighted corrected uniform	156/10000	1761/8239	30	66.2% [66.1%, 66.4%]	65.8% [65.7%, 65.9%]	0.4 [0.3, 0.6]
FD002 hybrid unweighted uniform	156/259	42/217	30	65.6% [64.5%, 66.7%]	64.9% [64.6%, 65.3%]	0.6 [-0.5, 1.8]
FD002 hybrid unweighted uniform	156/10000	1761/8239	30	65.5% [65.4%, 65.7%]	65.2% [65.1%, 65.3%]	0.3 [0.1, 0.5]
FD002 covariate weighted raw final	156/259	53/206	200	15.5% [12.5%, 18.5%]	11.7% [10.2%, 13.3%]	3.7 [0.5, 7.2]
FD002 covariate weighted raw final	156/10000	2052/7948	200	14.8% [14.3%, 15.3%]	11.5% [11.3%, 11.8%]	3.3 [2.8, 3.8]
FD002 covariate weighted corrected final	156/259	53/206	200	11.7% [9.1%, 14.4%]	8.8% [7.3%, 10.4%]	2.8 [-0.2, 6.0]
FD002 covariate weighted corrected final	156/10000	2052/7948	200	12.1% [11.7%, 12.6%]	9.1% [8.9%, 9.4%]	3.0 [2.5, 3.5]
FD002 covariate unweighted final	156/259	53/206	200	11.5% [9.1%, 13.9%]	8.3% [7.0%, 9.7%]	3.2 [0.5, 6.0]
FD002 covariate unweighted final	156/10000	2052/7948	200	11.3% [10.9%, 11.7%]	8.5% [8.2%, 8.7%]	2.8 [2.4, 3.3]
FD002 covariate weighted raw uniform	156/259	42/217	200	15.4% [10.9%, 20.1%]	15.6% [13.4%, 17.7%]	-0.2 [-5.2, 5.1]
FD002 covariate weighted raw uniform	156/10000	1761/8239	200	15.2% [14.5%, 15.9%]	16.4% [16.0%, 16.7%]	-1.2 [-2.0, -0.4]
FD002 covariate weighted corrected uniform	156/259	42/217	200	9.1% [6.3%, 12.0%]	9.0% [7.6%, 10.2%]	0.1 [-3.0, 3.4]
FD002 covariate weighted corrected uniform	156/10000	1761/8239	200	8.7% [8.3%, 9.2%]	9.4% [9.2%, 9.7%]	-0.7 [-1.2, -0.3]
FD002 covariate unweighted uniform	156/259	42/217	200	10.5% [7.3%, 13.9%]	10.4% [8.9%, 11.9%]	0.1 [-3.5, 3.8]
FD002 covariate unweighted uniform	156/10000	1761/8239	200	10.1% [9.6%, 10.7%]	11.0% [10.7%, 11.2%]	-0.8 [-1.4, -0.3]
FD003 lstm weighted raw final	60/100	18/82	30	42.2% [35.7%, 49.0%]	42.8% [39.4%, 46.2%]	-0.5 [-8.0, 6.8]
FD003 lstm weighted raw final	60/10000	1998/8002	30	46.3% [45.7%, 46.9%]	42.0% [41.7%, 42.4%]	4.3 [3.5, 5.0]

Group / variant / policy	Fit / test engines	Band / elsewhere engines	Fits	Band rate [95% CI]	Elsewhere rate [95% CI]	Difference, pp [95% CI]
FD003 lstm weighted corrected final	60/100	18/82	30	18.0% [13.8%, 22.8%]	19.1% [16.7%, 21.6%]	-1.2 [-6.3, 4.2]
FD003 lstm weighted corrected final	60/10000	1998/8002	30	20.1% [19.7%, 20.6%]	17.7% [17.5%, 17.9%]	2.4 [1.9, 3.0]
FD003 lstm unweighted final	60/100	18/82	30	16.3% [12.9%, 20.0%]	16.5% [14.8%, 18.1%]	-0.2 [-3.9, 3.8]
FD003 lstm unweighted final	60/10000	1998/8002	30	17.6% [17.3%, 17.9%]	15.9% [15.7%, 16.0%]	1.7 [1.4, 2.1]
FD003 lstm weighted raw uniform	60/100	21/79	30	40.5% [34.6%, 46.4%]	40.0% [37.2%, 42.7%]	0.5 [-6.2, 7.3]
FD003 lstm weighted raw uniform	60/10000	1786/8214	30	38.6% [38.0%, 39.1%]	39.7% [39.4%, 40.0%]	-1.1 [-1.8, -0.5]
FD003 lstm weighted corrected uniform	60/100	21/79	30	14.9% [12.5%, 17.8%]	13.5% [12.6%, 14.6%]	1.4 [-1.3, 4.4]
FD003 lstm weighted corrected uniform	60/10000	1786/8214	30	13.1% [12.9%, 13.3%]	13.3% [13.2%, 13.4%]	-0.2 [-0.4, 0.0]
FD003 lstm unweighted uniform	60/100	21/79	30	15.2% [12.8%, 17.8%]	15.4% [14.2%, 16.7%]	-0.2 [-2.8, 2.7]
FD003 lstm unweighted uniform	60/10000	1786/8214	30	14.8% [14.5%, 15.0%]	15.4% [15.3%, 15.5%]	-0.6 [-0.9, -0.3]
FD003 hybrid weighted raw final	60/100	18/82	30	46.1% [42.4%, 50.0%]	47.2% [45.3%, 49.1%]	-1.0 [-5.3, 3.5]
FD003 hybrid weighted raw final	60/10000	1998/8002	30	48.2% [47.8%, 48.5%]	46.8% [46.6%, 47.0%]	1.3 [0.9, 1.7]
FD003 hybrid weighted corrected final	60/100	18/82	30	65.4% [57.8%, 72.9%]	65.7% [61.8%, 69.7%]	-0.4 [-9.4, 8.1]
FD003 hybrid weighted corrected final	60/10000	1998/8002	30	70.0% [69.3%, 70.7%]	65.3% [64.9%, 65.7%]	4.7 [3.9, 5.5]
FD003 hybrid unweighted final	60/100	18/82	30	69.3% [60.5%, 77.7%]	69.9% [65.9%, 74.1%]	-0.7 [-10.5, 8.3]
FD003 hybrid unweighted final	60/10000	1998/8002	30	73.6% [72.9%, 74.4%]	69.3% [68.9%, 69.7%]	4.4 [3.5, 5.2]
FD003 hybrid weighted raw uniform	60/100	21/79	30	45.2% [42.9%, 47.6%]	43.0% [41.6%, 44.4%]	2.2 [-0.6, 4.9]
FD003 hybrid weighted raw uniform	60/10000	1786/8214	30	43.0% [42.7%, 43.3%]	43.0% [42.9%, 43.1%]	0.0 [-0.3, 0.3]
FD003 hybrid weighted corrected uniform	60/100	21/79	30	62.2% [56.3%, 68.2%]	60.8% [58.0%, 63.6%]	1.4 [-5.2, 8.0]
FD003 hybrid weighted corrected uniform	60/10000	1786/8214	30	59.2% [58.6%, 59.8%]	60.1% [59.8%, 60.3%]	-0.9 [-1.5, -0.3]
FD003 hybrid unweighted uniform	60/100	21/79	30	60.5% [55.4%, 66.4%]	59.6% [57.3%, 62.2%]	0.9 [-4.8, 7.3]
FD003 hybrid unweighted uniform	60/10000	1786/8214	30	58.6% [58.1%, 59.0%]	58.6% [58.4%, 58.8%]	-0.0 [-0.6, 0.5]

Group / variant / policy	Fit / test engines	Band / elsewhere engines	Fits	Band rate [95% CI]	Elsewhere rate [95% CI]	Difference, pp [95% CI]
FD003 covariate weighted raw final	60/100	18/82	200	36.4% [21.0%, 52.1%]	35.4% [28.6%, 42.2%]	1.1 [-15.7, 17.3]
FD003 covariate weighted raw final	60/10000	1998/8002	200	44.3% [42.9%, 45.7%]	35.3% [34.6%, 36.0%]	9.0 [7.3, 10.5]
FD003 covariate weighted corrected final	60/100	18/82	200	17.6% [9.4%, 25.8%]	16.9% [13.2%, 20.5%]	0.7 [-8.3, 9.3]
FD003 covariate weighted corrected final	60/10000	1998/8002	200	21.4% [20.7%, 22.2%]	16.5% [16.1%, 16.9%]	4.9 [4.0, 5.7]
FD003 covariate unweighted final	60/100	18/82	200	18.9% [10.2%, 27.5%]	18.3% [14.3%, 22.3%]	0.5 [-9.2, 9.7]
FD003 covariate unweighted final	60/10000	1998/8002	200	23.1% [22.3%, 23.9%]	17.8% [17.4%, 18.2%]	5.3 [4.4, 6.2]
FD003 covariate weighted raw uniform	60/100	21/79	200	35.7% [21.5%, 49.9%]	35.0% [27.9%, 41.6%]	0.7 [-15.1, 16.6]
FD003 covariate weighted raw uniform	60/10000	1786/8214	200	31.9% [30.4%, 33.4%]	35.3% [34.7%, 36.0%]	-3.4 [-5.1, -1.8]
FD003 covariate weighted corrected uniform	60/100	21/79	200	17.4% [9.9%, 24.6%]	17.0% [13.3%, 20.5%]	0.4 [-7.8, 8.5]
FD003 covariate weighted corrected uniform	60/10000	1786/8214	200	15.3% [14.5%, 16.1%]	17.0% [16.6%, 17.4%]	-1.7 [-2.6, -0.8]
FD003 covariate unweighted uniform	60/100	21/79	200	18.5% [10.6%, 26.3%]	18.0% [14.1%, 21.8%]	0.5 [-8.2, 9.2]
FD003 covariate unweighted uniform	60/10000	1786/8214	200	16.3% [15.5%, 17.2%]	18.1% [17.7%, 18.4%]	-1.7 [-2.6, -0.8]
FD004 lstm weighted raw final	149/248	50/198	30	36.9% [29.9%, 44.0%]	27.6% [24.2%, 31.4%]	9.3 [1.0, 17.0]
FD004 lstm weighted raw final	149/10000	1940/8060	30	34.8% [33.7%, 35.9%]	28.1% [27.5%, 28.7%]	6.7 [5.5, 8.0]
FD004 lstm weighted corrected final	149/248	50/198	30	21.9% [16.7%, 27.2%]	15.8% [13.3%, 18.5%]	6.2 [0.1, 12.0]
FD004 lstm weighted corrected final	149/10000	1940/8060	30	21.5% [20.7%, 22.3%]	16.6% [16.1%, 17.0%]	4.9 [4.0, 5.9]
FD004 lstm unweighted final	149/248	50/198	30	20.5% [16.0%, 25.0%]	16.5% [14.4%, 18.9%]	4.0 [-1.1, 9.0]
FD004 lstm unweighted final	149/10000	1940/8060	30	21.1% [20.4%, 21.9%]	17.0% [16.7%, 17.4%]	4.1 [3.3, 4.9]
FD004 lstm weighted raw uniform	149/248	44/204	30	31.8% [23.3%, 40.8%]	28.6% [24.4%, 32.3%]	3.2 [-5.9, 12.8]
FD004 lstm weighted raw uniform	149/10000	1823/8177	30	28.0% [26.7%, 29.4%]	29.5% [28.9%, 30.2%]	-1.5 [-2.9, 0.0]
FD004 lstm weighted corrected uniform	149/248	44/204	30	19.5% [14.1%, 25.2%]	16.8% [14.2%, 19.1%]	2.7 [-3.2, 9.0]
FD004 lstm weighted corrected uniform	149/10000	1823/8177	30	17.2% [16.3%, 18.1%]	17.5% [17.1%, 17.9%]	-0.3 [-1.3, 0.6]

Group / variant / policy	Fit / test engines	Band / elsewhere engines	Fits	Band rate [95% CI]	Elsewhere rate [95% CI]	Difference, pp [95% CI]
FD004 lstm unweighted uniform	149/248	44/204	30	24.2% [18.4%, 30.3%]	21.8% [19.1%, 24.3%]	2.4 [-3.9, 8.9]
FD004 lstm unweighted uniform	149/10000	1823/8177	30	21.5% [20.6%, 22.4%]	22.6% [22.2%, 23.0%]	-1.1 [-2.1, -0.1]
FD004 hybrid weighted raw final	149/248	50/198	30	45.4% [44.3%, 46.6%]	44.8% [44.3%, 45.4%]	0.6 [-0.6, 1.8]
FD004 hybrid weighted raw final	149/10000	1940/8060	30	45.0% [44.8%, 45.2%]	44.7% [44.6%, 44.8%]	0.3 [0.1, 0.5]
FD004 hybrid weighted corrected final	149/248	50/198	30	57.7% [53.5%, 62.0%]	52.1% [50.1%, 54.2%]	5.7 [0.9, 10.3]
FD004 hybrid weighted corrected final	149/10000	1940/8060	30	55.9% [55.3%, 56.6%]	52.4% [52.1%, 52.7%]	3.6 [2.8, 4.3]
FD004 hybrid unweighted final	149/248	50/198	30	61.4% [57.2%, 65.7%]	55.0% [53.0%, 57.1%]	6.4 [1.5, 11.0]
FD004 hybrid unweighted final	149/10000	1940/8060	30	59.3% [58.6%, 59.9%]	55.8% [55.4%, 56.1%]	3.5 [2.8, 4.3]
FD004 hybrid weighted raw uniform	149/248	44/204	30	46.1% [44.6%, 47.5%]	46.0% [45.4%, 46.7%]	0.0 [-1.6, 1.6]
FD004 hybrid weighted raw uniform	149/10000	1823/8177	30	46.0% [45.8%, 46.3%]	46.3% [46.2%, 46.4%]	-0.3 [-0.5, -0.0]
FD004 hybrid weighted corrected uniform	149/248	44/204	30	46.6% [44.6%, 48.7%]	45.3% [44.5%, 46.2%]	1.3 [-0.9, 3.6]
FD004 hybrid weighted corrected uniform	149/10000	1823/8177	30	45.6% [45.2%, 45.9%]	45.5% [45.3%, 45.6%]	0.1 [-0.3, 0.4]
FD004 hybrid unweighted uniform	149/248	44/204	30	47.5% [45.8%, 49.5%]	46.5% [45.8%, 47.3%]	1.0 [-0.9, 3.1]
FD004 hybrid unweighted uniform	149/10000	1823/8177	30	47.1% [46.8%, 47.4%]	46.4% [46.3%, 46.6%]	0.6 [0.3, 0.9]
FD004 covariate weighted raw final	149/248	50/198	200	17.2% [13.7%, 20.6%]	14.5% [12.7%, 16.6%]	2.6 [-1.4, 6.5]
FD004 covariate weighted raw final	149/10000	1940/8060	200	16.0% [15.4%, 16.6%]	14.1% [13.8%, 14.3%]	1.9 [1.3, 2.6]
FD004 covariate weighted corrected final	149/248	50/198	200	14.9% [11.5%, 18.3%]	10.6% [9.1%, 12.4%]	4.3 [0.4, 8.1]
FD004 covariate weighted corrected final	149/10000	1940/8060	200	13.6% [13.0%, 14.1%]	10.8% [10.5%, 11.1%]	2.8 [2.2, 3.4]
FD004 covariate unweighted final	149/248	50/198	200	13.5% [10.4%, 16.5%]	9.8% [8.4%, 11.5%]	3.6 [0.1, 7.1]
FD004 covariate unweighted final	149/10000	1940/8060	200	12.4% [11.9%, 12.8%]	9.9% [9.6%, 10.1%]	2.5 [1.9, 3.0]
FD004 covariate weighted raw uniform	149/248	44/204	200	15.7% [11.4%, 20.4%]	16.1% [13.7%, 18.2%]	-0.3 [-5.0, 4.6]
FD004 covariate weighted raw uniform	149/10000	1823/8177	200	14.5% [13.8%, 15.2%]	16.5% [16.2%, 16.9%]	-2.0 [-2.8, -1.2]

Group / variant / policy	Fit / test engines	Band / elsewhere engines	Fits	Band rate [95% CI]	Elsewhere rate [95% CI]	Difference, pp [95% CI]
FD004 covariate weighted corrected uniform	149/248	44/204	200	9.2% [6.5%, 11.9%]	9.2% [7.8%, 10.5%]	-0.0 [-2.8, 3.0]
FD004 covariate weighted corrected uniform	149/10000	1823/8177	200	8.5% [8.1%, 8.9%]	9.6% [9.3%, 9.8%]	-1.1 [-1.5, -0.6]
FD004 covariate unweighted uniform	149/248	44/204	200	10.3% [7.2%, 13.4%]	10.2% [8.7%, 11.7%]	0.1 [-3.1, 3.4]
FD004 covariate unweighted uniform	149/10000	1823/8177	200	9.4% [8.9%, 9.9%]	10.6% [10.4%, 10.9%]	-1.2 [-1.7, -0.7]

Table S11. Full non-collapsed pipeline-check percentiles and real rates

Source: `null_distribution.csv`; all rates are percentages. These are descriptive pipeline-check values, not a real-data reference benchmark.

Group	Fit / pipeline / real engines	Pipeline fits	Real rate	Min	2.5	25	50	75	97.5	Max	Location
FD001 lstm	60/10000/100	22	20.4%	0.0%	0.0%	0.3%	99.7%	100.0%	100.0%	100.0%	inside
FD001 hybrid	60/10000/100	5	13.6%	10.7%	14.3%	46.6%	49.8%	50.3%	55.2%	55.8%	below
FD002 lstm	156/10000/259	30	9.6%	0.7%	1.0%	5.8%	25.5%	47.8%	57.5%	76.9%	inside
FD002 hybrid	156/10000/259	10	8.9%	0.0%	0.8%	7.9%	18.4%	26.2%	88.8%	100.0%	inside
FD003 lstm	60/10000/100	28	12.8%	0.0%	0.0%	6.4%	37.6%	58.7%	100.0%	100.0%	inside
FD003 hybrid	60/10000/100	17	11.8%	1.3%	7.1%	22.1%	31.2%	42.2%	79.3%	97.0%	inside
FD004 lstm	149/10000/248	30	16.6%	3.9%	4.3%	12.8%	30.2%	43.8%	60.2%	64.0%	inside
FD004 hybrid	149/10000/248	18	12.7%	2.0%	3.6%	10.1%	22.8%	41.0%	94.5%	95.5%	inside

Table S12. Per-fit difficulty: all retained synthetic and real fits

Source: `difficulty_per_fit.csv`. Rows: 200. The full source is in `validation_data`.

Group / cohort	Fit ID	Fit / test engines	Capped RMSE	AUC	Event base rate	Median raw width
FD001 lstm synthetic noncollapsed	1	60/10000	45.4	0.651377	41.5%	36.8
FD001 lstm synthetic noncollapsed	3	60/10000	47.7	0.636140	41.5%	39.8
FD001 lstm synthetic noncollapsed	4	60/10000	46.3	0.575869	41.5%	67.6
FD001 lstm synthetic noncollapsed	5	60/10000	46.0	0.643054	41.5%	42.4
FD001 lstm synthetic noncollapsed	6	60/10000	42.9	0.629793	41.5%	89.4
FD001 lstm synthetic noncollapsed	7	60/10000	45.3	0.365872	41.5%	43.7
FD001 lstm synthetic noncollapsed	8	60/10000	42.8	0.645371	41.5%	83.4
FD001 lstm synthetic noncollapsed	10	60/10000	48.0	0.654428	41.5%	42.0
FD001 lstm synthetic noncollapsed	11	60/10000	43.9	0.653568	41.5%	62.4
FD001 lstm synthetic noncollapsed	13	60/10000	43.5	0.623041	41.5%	102
FD001 lstm synthetic noncollapsed	14	60/10000	47.4	0.634451	41.5%	39.3
FD001 lstm synthetic noncollapsed	15	60/10000	42.8	0.526458	41.5%	69.5
FD001 lstm synthetic noncollapsed	16	60/10000	47.2	0.649005	41.5%	43.0
FD001 lstm synthetic noncollapsed	17	60/10000	48.1	0.349689	41.5%	34.1
FD001 lstm synthetic noncollapsed	19	60/10000	42.0	0.653508	41.5%	89.6
FD001 lstm synthetic noncollapsed	20	60/10000	43.0	0.633582	41.5%	83.9
FD001 lstm synthetic noncollapsed	21	60/10000	47.1	0.620804	41.5%	46.9
FD001 lstm synthetic noncollapsed	24	60/10000	49.5	0.653424	41.5%	48.2
FD001 lstm synthetic noncollapsed	25	60/10000	50.0	0.588560	41.5%	24.6

Group / cohort	Fit ID	Fit / test engines	Capped RMSE	AUC	Event base rate	Median raw width
FD001 lstm synthetic noncollapsed	27	60/10000	42.5	0.656526	41.5%	74.6
FD001 lstm synthetic noncollapsed	28	60/10000	44.5	0.554706	41.5%	48.6
FD001 lstm synthetic noncollapsed	29	60/10000	41.6	0.653305	41.5%	56.7
FD001 lstm real	1101	60/100	13.7	0.986667	25.0%	66.3
FD001 lstm real	1202	60/100	14.6	0.986667	25.0%	83.1
FD001 lstm real	1303	60/100	14.9	0.985600	25.0%	77.4
FD001 lstm real	1404	60/100	15.4	0.982400	25.0%	68.2
FD001 lstm real	1505	60/100	14.9	0.989867	25.0%	68.7
FD001 hybrid synthetic noncollapsed	9	60/10000	46.1	0.353316	41.5%	119
FD001 hybrid synthetic noncollapsed	14	60/10000	42.7	0.643639	41.5%	73.4
FD001 hybrid synthetic noncollapsed	16	60/10000	40.5	0.651309	41.5%	122
FD001 hybrid synthetic noncollapsed	17	60/10000	42.1	0.654745	41.5%	107
FD001 hybrid synthetic noncollapsed	25	60/10000	43.0	0.651765	41.5%	112
FD001 hybrid real	1101	60/100	15.3	0.993600	25.0%	51.9
FD001 hybrid real	1202	60/100	18.1	0.988800	25.0%	54.5
FD001 hybrid real	1303	60/100	15.0	0.989333	25.0%	46.1
FD001 hybrid real	1404	60/100	15.8	0.985067	25.0%	50.3
FD001 hybrid real	1505	60/100	15.3	0.988800	25.0%	47.4
FD002 lstm synthetic noncollapsed	0	156/10000	41.7	0.596202	42.0%	131
FD002 lstm synthetic noncollapsed	1	156/10000	41.0	0.624169	42.0%	124
FD002 lstm synthetic noncollapsed	2	156/10000	41.1	0.649220	42.0%	105
FD002 lstm synthetic noncollapsed	3	156/10000	40.1	0.649437	42.0%	99.7
FD002 lstm synthetic noncollapsed	4	156/10000	39.7	0.647219	42.0%	127
FD002 lstm synthetic noncollapsed	5	156/10000	40.0	0.645868	42.0%	109
FD002 lstm synthetic noncollapsed	6	156/10000	40.4	0.649271	42.0%	87.6

Group / cohort	Fit ID	Fit / test engines	Capped RMSE	AUC	Event base rate	Median raw width
FD002 lstm synthetic noncollapsed	7	156/10000	44.0	0.648324	42.0%	93.5
FD002 lstm synthetic noncollapsed	8	156/10000	41.3	0.651367	42.0%	92.4
FD002 lstm synthetic noncollapsed	9	156/10000	40.2	0.642646	42.0%	129
FD002 lstm synthetic noncollapsed	10	156/10000	39.6	0.647341	42.0%	102
FD002 lstm synthetic noncollapsed	11	156/10000	45.9	0.643299	42.0%	130
FD002 lstm synthetic noncollapsed	12	156/10000	41.3	0.647874	42.0%	130
FD002 lstm synthetic noncollapsed	13	156/10000	41.5	0.648776	42.0%	88.5
FD002 lstm synthetic noncollapsed	14	156/10000	40.4	0.649796	42.0%	97.0
FD002 lstm synthetic noncollapsed	15	156/10000	40.9	0.650887	42.0%	106
FD002 lstm synthetic noncollapsed	16	156/10000	39.4	0.647913	42.0%	100
FD002 lstm synthetic noncollapsed	17	156/10000	39.7	0.650548	42.0%	89.2
FD002 lstm synthetic noncollapsed	18	156/10000	43.1	0.648572	42.0%	130
FD002 lstm synthetic noncollapsed	19	156/10000	41.1	0.645755	42.0%	123
FD002 lstm synthetic noncollapsed	20	156/10000	39.8	0.647266	42.0%	103
FD002 lstm synthetic noncollapsed	21	156/10000	41.4	0.651885	42.0%	114
FD002 lstm synthetic noncollapsed	22	156/10000	39.7	0.648628	42.0%	126
FD002 lstm synthetic noncollapsed	23	156/10000	40.4	0.649780	42.0%	87.4
FD002 lstm synthetic noncollapsed	24	156/10000	42.0	0.620205	42.0%	128
FD002 lstm synthetic noncollapsed	25	156/10000	43.9	0.626810	42.0%	123
FD002 lstm synthetic noncollapsed	26	156/10000	40.3	0.649071	42.0%	90.5

Group / cohort	Fit ID	Fit / test engines	Capped RMSE	AUC	Event base rate	Median raw width
FD002 lstm synthetic noncollapsed	27	156/10000	39.5	0.649668	42.0%	123
FD002 lstm synthetic noncollapsed	28	156/10000	40.5	0.649401	42.0%	111
FD002 lstm synthetic noncollapsed	29	156/10000	39.4	0.648729	42.0%	110
FD002 lstm real	1101	156/259	13.6	0.997516	23.6%	53.7
FD002 lstm real	1202	156/259	14.2	0.998013	23.6%	53.5
FD002 lstm real	1303	156/259	15.1	0.998427	23.6%	56.6
FD002 lstm real	1404	156/259	14.4	0.999089	23.6%	55.9
FD002 lstm real	1505	156/259	14.3	0.999006	23.6%	54.5
FD002 hybrid synthetic noncollapsed	3	156/10000	40.3	0.647316	42.0%	88.4
FD002 hybrid synthetic noncollapsed	7	156/10000	40.2	0.647343	42.0%	60.3
FD002 hybrid synthetic noncollapsed	18	156/10000	42.2	0.645214	42.0%	119
FD002 hybrid synthetic noncollapsed	22	156/10000	39.5	0.648462	42.0%	84.3
FD002 hybrid synthetic noncollapsed	23	156/10000	40.3	0.650413	42.0%	62.2
FD002 hybrid synthetic noncollapsed	24	156/10000	42.2	0.603965	42.0%	114
FD002 hybrid synthetic noncollapsed	25	156/10000	40.4	0.649820	42.0%	110
FD002 hybrid synthetic noncollapsed	26	156/10000	39.9	0.649607	42.0%	105
FD002 hybrid synthetic noncollapsed	28	156/10000	41.7	0.648452	42.0%	91.1
FD002 hybrid synthetic noncollapsed	29	156/10000	39.9	0.646238	42.0%	95.7
FD002 hybrid real	1101	156/259	15.0	0.998261	23.6%	56.6
FD002 hybrid real	1202	156/259	14.5	0.997268	23.6%	51.2
FD002 hybrid real	1303	156/259	15.3	0.998510	23.6%	54.3
FD002 hybrid real	1404	156/259	14.3	0.997019	23.6%	56.1
FD002 hybrid real	1505	156/259	17.9	0.991389	23.6%	53.0
FD003 lstm synthetic noncollapsed	0	60/10000	41.6	0.653232	40.8%	124
FD003 lstm synthetic noncollapsed	1	60/10000	42.7	0.605508	40.8%	137

Group / cohort	Fit ID	Fit / test engines	Capped RMSE	AUC	Event base rate	Median raw width
FD003 lstm synthetic noncollapsed	2	60/10000	44.1	0.367174	40.8%	124
FD003 lstm synthetic noncollapsed	3	60/10000	39.9	0.654099	40.8%	114
FD003 lstm synthetic noncollapsed	4	60/10000	42.2	0.653868	40.8%	130
FD003 lstm synthetic noncollapsed	5	60/10000	42.3	0.648748	40.8%	125
FD003 lstm synthetic noncollapsed	6	60/10000	41.3	0.651551	40.8%	117
FD003 lstm synthetic noncollapsed	7	60/10000	40.7	0.656477	40.8%	132
FD003 lstm synthetic noncollapsed	8	60/10000	40.7	0.640693	40.8%	129
FD003 lstm synthetic noncollapsed	9	60/10000	39.7	0.647266	40.8%	129
FD003 lstm synthetic noncollapsed	10	60/10000	40.5	0.654240	40.8%	111
FD003 lstm synthetic noncollapsed	11	60/10000	39.9	0.656205	40.8%	128
FD003 lstm synthetic noncollapsed	12	60/10000	44.1	0.373762	40.8%	126
FD003 lstm synthetic noncollapsed	14	60/10000	39.3	0.655992	40.8%	136
FD003 lstm synthetic noncollapsed	15	60/10000	41.1	0.654917	40.8%	133
FD003 lstm synthetic noncollapsed	16	60/10000	40.6	0.654715	40.8%	90.3
FD003 lstm synthetic noncollapsed	17	60/10000	45.3	0.654830	40.8%	115
FD003 lstm synthetic noncollapsed	18	60/10000	39.9	0.655229	40.8%	119
FD003 lstm synthetic noncollapsed	19	60/10000	43.7	0.622473	40.8%	129
FD003 lstm synthetic noncollapsed	20	60/10000	40.3	0.654709	40.8%	130
FD003 lstm synthetic noncollapsed	21	60/10000	40.1	0.655043	40.8%	108
FD003 lstm synthetic noncollapsed	22	60/10000	40.5	0.618995	40.8%	129

Group / cohort	Fit ID	Fit / test engines	Capped RMSE	AUC	Event base rate	Median raw width
FD003 lstm synthetic noncollapsed	23	60/10000	40.5	0.655291	40.8%	117
FD003 lstm synthetic noncollapsed	24	60/10000	40.9	0.655598	40.8%	121
FD003 lstm synthetic noncollapsed	26	60/10000	42.8	0.629315	40.8%	132
FD003 lstm synthetic noncollapsed	27	60/10000	42.2	0.656059	40.8%	125
FD003 lstm synthetic noncollapsed	28	60/10000	50.1	0.379198	40.8%	129
FD003 lstm synthetic noncollapsed	29	60/10000	41.5	0.656169	40.8%	124
FD003 lstm real	1101	60/100	17.6	0.990625	20.0%	49.3
FD003 lstm real	1202	60/100	16.3	0.994375	20.0%	42.7
FD003 lstm real	1303	60/100	17.8	0.997500	20.0%	46.0
FD003 lstm real	1404	60/100	18.7	0.995000	20.0%	45.1
FD003 lstm real	1505	60/100	19.1	0.990625	20.0%	50.9
FD003 hybrid synthetic noncollapsed	2	60/10000	43.6	0.631133	40.8%	102
FD003 hybrid synthetic noncollapsed	3	60/10000	44.4	0.643094	40.8%	96.7
FD003 hybrid synthetic noncollapsed	4	60/10000	48.0	0.627756	40.8%	67.2
FD003 hybrid synthetic noncollapsed	7	60/10000	42.1	0.650906	40.8%	64.0
FD003 hybrid synthetic noncollapsed	10	60/10000	43.8	0.654172	40.8%	54.5
FD003 hybrid synthetic noncollapsed	15	60/10000	42.0	0.656168	40.8%	81.8
FD003 hybrid synthetic noncollapsed	16	60/10000	45.6	0.656474	40.8%	101
FD003 hybrid synthetic noncollapsed	17	60/10000	46.8	0.635551	40.8%	54.0
FD003 hybrid synthetic noncollapsed	19	60/10000	49.6	0.547727	40.8%	112
FD003 hybrid synthetic noncollapsed	20	60/10000	40.1	0.653901	40.8%	129
FD003 hybrid synthetic noncollapsed	21	60/10000	45.8	0.653898	40.8%	83.0
FD003 hybrid synthetic noncollapsed	22	60/10000	39.4	0.656982	40.8%	128

Group / cohort	Fit ID	Fit / test engines	Capped RMSE	AUC	Event base rate	Median raw width
FD003 hybrid synthetic noncollapsed	23	60/10000	52.1	0.650920	40.8%	38.9
FD003 hybrid synthetic noncollapsed	24	60/10000	41.6	0.654955	40.8%	67.5
FD003 hybrid synthetic noncollapsed	26	60/10000	40.8	0.654610	40.8%	105
FD003 hybrid synthetic noncollapsed	27	60/10000	42.5	0.639832	40.8%	90.9
FD003 hybrid synthetic noncollapsed	29	60/10000	44.1	0.654791	40.8%	58.9
FD003 hybrid real	1101	60/100	15.9	1.000000	20.0%	34.9
FD003 hybrid real	1202	60/100	14.1	0.997500	20.0%	31.7
FD003 hybrid real	1303	60/100	16.1	0.996875	20.0%	30.4
FD003 hybrid real	1404	60/100	14.9	0.996875	20.0%	29.3
FD003 hybrid real	1505	60/100	15.3	0.993125	20.0%	30.0
FD004 lstm synthetic noncollapsed	0	149/10000	42.5	0.654165	40.3%	103
FD004 lstm synthetic noncollapsed	1	149/10000	39.2	0.654963	40.3%	106
FD004 lstm synthetic noncollapsed	2	149/10000	40.8	0.648706	40.3%	97.2
FD004 lstm synthetic noncollapsed	3	149/10000	40.3	0.657896	40.3%	102
FD004 lstm synthetic noncollapsed	4	149/10000	41.3	0.648631	40.3%	85.9
FD004 lstm synthetic noncollapsed	5	149/10000	40.1	0.641142	40.3%	132
FD004 lstm synthetic noncollapsed	6	149/10000	39.1	0.657810	40.3%	110
FD004 lstm synthetic noncollapsed	7	149/10000	40.8	0.630992	40.3%	130
FD004 lstm synthetic noncollapsed	8	149/10000	41.0	0.659329	40.3%	101
FD004 lstm synthetic noncollapsed	9	149/10000	39.8	0.648891	40.3%	116
FD004 lstm synthetic noncollapsed	10	149/10000	40.2	0.651344	40.3%	128
FD004 lstm synthetic noncollapsed	11	149/10000	40.8	0.653304	40.3%	123

Group / cohort	Fit ID	Fit / test engines	Capped RMSE	AUC	Event base rate	Median raw width
FD004 lstm synthetic noncollapsed	12	149/10000	39.5	0.657494	40.3%	116
FD004 lstm synthetic noncollapsed	13	149/10000	42.9	0.634775	40.3%	126
FD004 lstm synthetic noncollapsed	14	149/10000	39.6	0.656848	40.3%	110
FD004 lstm synthetic noncollapsed	15	149/10000	41.8	0.654296	40.3%	117
FD004 lstm synthetic noncollapsed	16	149/10000	39.5	0.647248	40.3%	93.6
FD004 lstm synthetic noncollapsed	17	149/10000	42.1	0.656949	40.3%	110
FD004 lstm synthetic noncollapsed	18	149/10000	41.4	0.632280	40.3%	128
FD004 lstm synthetic noncollapsed	19	149/10000	40.7	0.657172	40.3%	112
FD004 lstm synthetic noncollapsed	20	149/10000	41.5	0.637755	40.3%	129
FD004 lstm synthetic noncollapsed	21	149/10000	41.1	0.647240	40.3%	122
FD004 lstm synthetic noncollapsed	22	149/10000	39.7	0.651512	40.3%	120
FD004 lstm synthetic noncollapsed	23	149/10000	39.9	0.654650	40.3%	123
FD004 lstm synthetic noncollapsed	24	149/10000	39.6	0.653477	40.3%	111
FD004 lstm synthetic noncollapsed	25	149/10000	42.7	0.656256	40.3%	94.0
FD004 lstm synthetic noncollapsed	26	149/10000	40.4	0.658399	40.3%	98.4
FD004 lstm synthetic noncollapsed	27	149/10000	40.6	0.654655	40.3%	106
FD004 lstm synthetic noncollapsed	28	149/10000	41.8	0.657068	40.3%	102
FD004 lstm synthetic noncollapsed	29	149/10000	45.4	0.653490	40.3%	130
FD004 lstm real	1101	149/248	16.9	0.989453	21.4%	47.1
FD004 lstm real	1202	149/248	16.2	0.987325	21.4%	46.0
FD004 lstm real	1303	149/248	16.2	0.990227	21.4%	44.1
FD004 lstm real	1404	149/248	16.9	0.988582	21.4%	46.1
FD004 lstm real	1505	149/248	16.6	0.987712	21.4%	48.9

Group / cohort	Fit ID	Fit / test engines	Capped RMSE	AUC	Event base rate	Median raw width
FD004 hybrid synthetic noncollapsed	0	149/10000	41.2	0.655310	40.3%	78.6
FD004 hybrid synthetic noncollapsed	1	149/10000	39.6	0.651249	40.3%	104
FD004 hybrid synthetic noncollapsed	2	149/10000	40.0	0.654923	40.3%	91.3
FD004 hybrid synthetic noncollapsed	3	149/10000	40.5	0.398778	40.3%	103
FD004 hybrid synthetic noncollapsed	5	149/10000	39.5	0.655795	40.3%	114
FD004 hybrid synthetic noncollapsed	9	149/10000	39.1	0.649828	40.3%	87.0
FD004 hybrid synthetic noncollapsed	10	149/10000	40.2	0.497859	40.3%	105
FD004 hybrid synthetic noncollapsed	14	149/10000	39.4	0.650692	40.3%	97.2
FD004 hybrid synthetic noncollapsed	16	149/10000	39.6	0.370025	40.3%	103
FD004 hybrid synthetic noncollapsed	17	149/10000	43.7	0.657076	40.3%	77.9
FD004 hybrid synthetic noncollapsed	18	149/10000	41.8	0.653260	40.3%	93.1
FD004 hybrid synthetic noncollapsed	19	149/10000	41.7	0.656687	40.3%	98.8
FD004 hybrid synthetic noncollapsed	20	149/10000	39.6	0.650846	40.3%	112
FD004 hybrid synthetic noncollapsed	21	149/10000	42.8	0.653903	40.3%	109
FD004 hybrid synthetic noncollapsed	22	149/10000	39.8	0.462007	40.3%	108
FD004 hybrid synthetic noncollapsed	23	149/10000	39.4	0.654686	40.3%	83.2
FD004 hybrid synthetic noncollapsed	25	149/10000	43.5	0.654073	40.3%	91.0
FD004 hybrid synthetic noncollapsed	29	149/10000	41.9	0.471059	40.3%	116
FD004 hybrid real	1101	149/248	16.8	0.991388	21.4%	45.0
FD004 hybrid real	1202	149/248	16.3	0.983745	21.4%	38.9
FD004 hybrid real	1303	149/248	17.0	0.989550	21.4%	47.9
FD004 hybrid real	1404	149/248	16.3	0.982003	21.4%	43.8

Group / cohort	Fit ID	Fit / test engines	Capped RMSE	AUC	Event base rate	Median raw width
FD004 hybrid real	1505	149/248	16.3	0.982777	21.4%	42.5

Source and uncertainty notes

Synthetic shape is 1, with scale(x)=60 exp(0.5x), x in {-1,+1}. Training lifetimes are 30 plus an exponential excess lifetime. Final-window residual RUL depends on x but is age-independent conditional on x; uniform-endpoint conditional RUL depends on age and x. Raw widths are unprojected. Bootstrap CIs resample engines, preserving fixed fits. The paired factorial ranges are not CIs. Negative results, empty strata and shared FD001/FD003 covariate states remain recorded in the source tables. Independent human review has not been performed.

Table S13a. Positive controls: every group, subset-combined and pooled estimate

Source: `A_controls.csv`. Rates are percentages; distances are probability units.

Group / scope / control	Fit / test engines	Fits	Rate [95% CI]	Distance [95% CI]
FD001 lstm group raw	60/100	5	20.4% [14.2%, 26.6%]	0.021 [0.0129, 0.0301]
FD001 lstm group uniform	60/100	5	88.7% [85.9%, 91.2%]	0.401 [0.378, 0.421]
FD001 lstm group one minus p	60/100	5	92.0% [87.0%, 96.4%]	0.806 [0.75, 0.857]
FD001 lstm group logit plus 1	60/100	5	20.8% [14.0%, 28.4%]	0.0468 [0.0289, 0.0676]
FD001 lstm group logit minus 1	60/100	5	20.2% [13.8%, 27.2%]	0.0147 [0.00948, 0.0204]
FD001 hybrid group raw	60/100	5	13.6% [8.4%, 19.0%]	0.0138 [0.0079, 0.0206]
FD001 hybrid group uniform	60/100	5	89.2% [86.4%, 91.7%]	0.404 [0.382, 0.424]
FD001 hybrid group one minus p	60/100	5	97.2% [95.4%, 98.8%]	0.836 [0.789, 0.879]
FD001 hybrid group logit plus 1	60/100	5	18.6% [12.4%, 25.2%]	0.0283 [0.017, 0.0413]
FD001 hybrid group logit minus 1	60/100	5	12.0% [7.6%, 16.6%]	0.0115 [0.0052, 0.0195]
FD002 lstm group raw	156/259	5	9.6% [6.7%, 12.5%]	0.00671 [0.00416, 0.00952]
FD002 lstm group uniform	156/259	5	90.2% [88.5%, 91.6%]	0.412 [0.399, 0.424]
FD002 lstm group one minus p	156/259	5	97.3% [95.8%, 98.7%]	0.865 [0.839, 0.89]
FD002 lstm group logit plus 1	156/259	5	15.4% [11.5%, 19.4%]	0.0182 [0.0122, 0.0246]
FD002 lstm group logit minus 1	156/259	5	3.5% [2.2%, 4.9%]	0.0011 [0.000497, 0.00189]
FD002 hybrid group raw	156/259	5	8.9% [6.6%, 11.4%]	0.0072 [0.00431, 0.0104]
FD002 hybrid group uniform	156/259	5	89.3% [87.7%, 90.7%]	0.405 [0.392, 0.417]
FD002 hybrid group one minus p	156/259	5	97.1% [95.8%, 98.3%]	0.845 [0.818, 0.871]
FD002 hybrid group logit plus 1	156/259	5	14.9% [11.6%, 18.3%]	0.0192 [0.0131, 0.0257]
FD002 hybrid group logit minus 1	156/259	5	5.6% [3.9%, 7.2%]	0.00407 [0.00236, 0.00593]
FD003 lstm group raw	60/100	5	12.8% [8.2%, 18.0%]	0.015 [0.00787, 0.0237]
FD003 lstm group uniform	60/100	5	87.8% [84.8%, 90.5%]	0.393 [0.367, 0.415]
FD003 lstm group one minus p	60/100	5	95.8% [92.8%, 98.2%]	0.823 [0.771, 0.87]

Group / scope / control	Fit / test engines	Fits	Rate [95% CI]	Distance [95% CI]
FD003 lstm group logit plus 1	60/100	5	22.6% [15.8%, 30.2%]	0.0304 [0.0184, 0.0443]
FD003 lstm group logit minus 1	60/100	5	5.8% [3.2%, 9.0%]	0.00608 [0.0027, 0.0102]
FD003 hybrid group raw	60/100	5	11.8% [7.6%, 16.4%]	0.0149 [0.00814, 0.0223]
FD003 hybrid group uniform	60/100	5	89.6% [87.4%, 91.7%]	0.408 [0.389, 0.425]
FD003 hybrid group one minus p	60/100	5	96.6% [94.4%, 98.4%]	0.846 [0.801, 0.886]
FD003 hybrid group logit plus 1	60/100	5	17.4% [11.6%, 23.6%]	0.0311 [0.019, 0.0449]
FD003 hybrid group logit minus 1	60/100	5	8.2% [5.0%, 11.8%]	0.00738 [0.00391, 0.0116]
FD004 lstm group raw	149/248	5	16.6% [13.1%, 20.3%]	0.0137 [0.00992, 0.018]
FD004 lstm group uniform	149/248	5	86.7% [84.8%, 88.5%]	0.385 [0.37, 0.4]
FD004 lstm group one minus p	149/248	5	95.2% [93.1%, 96.9%]	0.81 [0.777, 0.841]
FD004 lstm group logit plus 1	149/248	5	24.0% [19.4%, 28.9%]	0.032 [0.0237, 0.041]
FD004 lstm group logit minus 1	149/248	5	8.5% [6.0%, 11.1%]	0.00383 [0.00245, 0.0054]
FD004 hybrid group raw	149/248	5	12.7% [9.8%, 15.6%]	0.00857 [0.00502, 0.0134]
FD004 hybrid group uniform	149/248	5	86.2% [84.4%, 88.1%]	0.38 [0.365, 0.394]
FD004 hybrid group one minus p	149/248	5	97.0% [95.3%, 98.4%]	0.804 [0.772, 0.834]
FD004 hybrid group logit plus 1	149/248	5	19.2% [15.5%, 23.1%]	0.0205 [0.0141, 0.0283]
FD004 hybrid group logit minus 1	149/248	5	9.0% [6.9%, 11.3%]	0.00451 [0.00258, 0.00717]
FD001 combined subset combined raw	60/100	10	17.0% [11.8%, 22.4%]	0.0174 [0.011, 0.0244]
FD001 combined subset combined uniform	60/100	10	88.9% [86.2%, 91.4%]	0.402 [0.38, 0.422]
FD001 combined subset combined one minus p	60/100	10	94.6% [91.3%, 97.5%]	0.821 [0.77, 0.868]
FD001 combined subset combined logit plus 1	60/100	10	19.7% [13.3%, 26.8%]	0.0376 [0.0237, 0.0536]
FD001 combined subset combined logit minus 1	60/100	10	16.1% [11.4%, 20.8%]	0.0131 [0.00833, 0.0187]
FD002 combined subset combined raw	156/259	10	9.2% [6.9%, 11.8%]	0.00695 [0.00469, 0.00944]
FD002 combined subset combined uniform	156/259	10	89.7% [88.2%, 91.1%]	0.408 [0.396, 0.42]
FD002 combined subset combined one minus p	156/259	10	97.2% [95.9%, 98.4%]	0.855 [0.829, 0.88]
FD002 combined subset combined logit plus 1	156/259	10	15.1% [11.8%, 18.5%]	0.0187 [0.0134, 0.0243]
FD002 combined subset combined logit minus 1	156/259	10	4.5% [3.3%, 5.8%]	0.00259 [0.00158, 0.00371]
FD003 combined subset combined raw	60/100	10	12.3% [8.4%, 16.6%]	0.015 [0.00894, 0.0219]
FD003 combined subset combined uniform	60/100	10	88.7% [86.1%, 91.0%]	0.4 [0.379, 0.419]
FD003 combined subset combined one minus p	60/100	10	96.2% [93.9%, 98.1%]	0.834 [0.789, 0.877]
FD003 combined subset combined logit plus 1	60/100	10	20.0% [14.1%, 26.4%]	0.0308 [0.0198, 0.0431]

Group / scope / control	Fit / test engines	Fits	Rate [95% CI]	Distance [95% CI]
FD003 combined subset combined logit minus 1	60/100	10	7.0% [4.5%, 9.9%]	0.00673 [0.00363, 0.0103]
FD004 combined subset combined raw	149/248	10	14.6% [11.8%, 17.8%]	0.0111 [0.008, 0.0151]
FD004 combined subset combined uniform	149/248	10	86.5% [84.6%, 88.2%]	0.383 [0.367, 0.397]
FD004 combined subset combined one minus p	149/248	10	96.1% [94.4%, 97.5%]	0.807 [0.775, 0.837]
FD004 combined subset combined logit plus 1	149/248	10	21.6% [17.7%, 25.7%]	0.0262 [0.0197, 0.0336]
FD004 combined subset combined logit minus 1	149/248	10	8.8% [6.7%, 10.8%]	0.00417 [0.0028, 0.00593]
FD001-FD004 combined pooled raw	425/707	40	12.7% [11.1%, 14.4%]	0.011 [0.00915, 0.0131]
FD001-FD004 combined pooled uniform	425/707	40	88.3% [87.3%, 89.3%]	0.397 [0.389, 0.405]
FD001-FD004 combined pooled one minus p	425/707	40	96.3% [95.4%, 97.2%]	0.831 [0.813, 0.847]
FD001-FD004 combined pooled logit plus 1	425/707	40	18.7% [16.5%, 21.1%]	0.0257 [0.0217, 0.0301]
FD001-FD004 combined pooled logit minus 1	425/707	40	8.0% [6.9%, 9.2%]	0.00522 [0.00419, 0.00635]

Table S13b. Projection: every test/calibration score and paired difference

Source: `B_projection.csv`. Rate changes are percentage points; other changes retain score units.

Group / role / metric	Fit / evaluation engines	Raw [CI]	Projected [CI]	Paired difference [CI]
FD001 lstm test Brier	60/100	0.0356 [0.0146, 0.0614]	0.0371 [0.0176, 0.0619]	0.00149 [-0.00668, 0.00964]
FD001 lstm test NLL	60/100	0.123 [0.0614, 0.201]	0.122 [0.0627, 0.198]	-0.000982 [-0.0193, 0.0177]
FD001 lstm test ECE5	60/100	0.0166 [0.0106, 0.0513]	0.0153 [0.0108, 0.0501]	-0.00132 [-0.0226, 0.0226]
FD001 lstm test ECE10	60/100	0.038 [0.032, 0.0757]	0.027 [0.0195, 0.0628]	-0.011 [-0.0344, 0.00546]
FD001 lstm test incompatibility rate	60/100	20.4% [14.2%, 26.6%]	0.0% [0.0%, 0.0%]	-20.4% [-26.6%, -14.2%]
FD001 lstm calibration Brier	60/20	0.0425 [0.000184, 0.124]	0.0408 [9.77e-05, 0.122]	-0.00172 [-0.00443, 0]
FD001 lstm calibration NLL	60/20	0.137 [0.00364, 0.386]	0.128 [0.00248, 0.374]	-0.0087 [-0.022, 0]
FD001 lstm calibration ECE5	60/20	0.0538 [0.00299, 0.151]	0.047 [0.000601, 0.14]	-0.0068 [-0.0185, 0.000902]
FD001 lstm calibration ECE10	60/20	0.0552 [0.00355, 0.151]	0.048 [0.00242, 0.141]	-0.00718 [-0.0192, 0.000157]
FD001 lstm calibration incompatibility rate	60/20	11.0% [0.0%, 23.0%]	0.0% [0.0%, 0.0%]	-11.0% [-23.0%, 0.0%]
FD001 hybrid test Brier	60/100	0.0348 [0.0131, 0.0625]	0.0326 [0.0122, 0.0591]	-0.00217 [-0.00689, 0.00197]
FD001 hybrid test NLL	60/100	0.121 [0.0484, 0.217]	0.113 [0.0438, 0.205]	-0.00855 [-0.0231, 0.0037]
FD001 hybrid test ECE5	60/100	0.0161 [0.00992, 0.0505]	0.0113 [0.00882, 0.0457]	-0.00481 [-0.0146, 0.0112]
FD001 hybrid test ECE10	60/100	0.0394 [0.0282, 0.0767]	0.029 [0.0214, 0.0637]	-0.0103 [-0.0187, -0.000511]
FD001 hybrid test incompatibility rate	60/100	13.6% [8.4%, 19.0%]	0.0% [0.0%, 0.0%]	-13.6% [-19.0%, -8.4%]
FD001 hybrid calibration Brier	60/20	0.0386 [3.44e-05, 0.114]	0.037 [3.36e-05, 0.111]	-0.00151 [-0.00391, 0]

Group / role / metric	Fit / evaluation engines	Raw [CI]	Projected [CI]	Paired difference [CI]
FD001 hybrid calibration NLL	60/20	0.116 [0.00146, 0.332]	0.11 [0.00145, 0.322]	-0.00625 [-0.0158, 0]
FD001 hybrid calibration ECE5	60/20	0.0511 [0.00144, 0.142]	0.0458 [0.00137, 0.133]	-0.00529 [-0.0136, 1.62e-05]
FD001 hybrid calibration ECE10	60/20	0.0516 [0.00144, 0.143]	0.0463 [0.00144, 0.134]	-0.00529 [-0.0136, 1.62e-05]
FD001 hybrid calibration incompatibility rate	60/20	6.0% [0.0%, 13.0%]	0.0% [0.0%, 0.0%]	-6.0% [-13.0%, 0.0%]
FD002 lstm test Brier	156/259	0.0251 [0.0124, 0.0402]	0.0212 [0.0105, 0.0344]	-0.00385 [-0.00698, -0.00112]
FD002 lstm test NLL	156/259	0.0816 [0.0425, 0.131]	0.0709 [0.038, 0.112]	-0.0108 [-0.0199, -0.0033]
FD002 lstm test ECE5	156/259	0.036 [0.0201, 0.0558]	0.0302 [0.0148, 0.0477]	-0.0058 [-0.00966, -0.00346]
FD002 lstm test ECE10	156/259	0.0364 [0.0226, 0.0563]	0.0304 [0.0188, 0.0485]	-0.00597 [-0.00891, -0.00271]
FD002 lstm test incompatibility rate	156/259	9.6% [6.7%, 12.5%]	0.0% [0.0%, 0.0%]	-9.6% [-12.5%, -6.7%]
FD002 lstm calibration Brier	156/52	0.00839 [0.000767, 0.0196]	0.00721 [0.000414, 0.0174]	-0.00118 [-0.00242, -0.000209]
FD002 lstm calibration NLL	156/52	0.0329 [0.00823, 0.0668]	0.0291 [0.00683, 0.0607]	-0.00374 [-0.00828, 9.41e-05]
FD002 lstm calibration ECE5	156/52	0.0238 [0.00476, 0.0495]	0.0191 [0.00524, 0.0444]	-0.0047 [-0.00823, 0.00109]
FD002 lstm calibration ECE10	156/52	0.0267 [0.00779, 0.052]	0.024 [0.00664, 0.0484]	-0.00273 [-0.00651, 0.000424]
FD002 lstm calibration incompatibility rate	156/52	10.8% [4.6%, 18.1%]	0.0% [0.0%, 0.0%]	-10.8% [-18.1%, -4.6%]
FD002 hybrid test Brier	156/259	0.0333 [0.0197, 0.0483]	0.0283 [0.016, 0.0417]	-0.00504 [-0.00864, -0.00194]
FD002 hybrid test NLL	156/259	0.118 [0.0706, 0.173]	0.0992 [0.0576, 0.15]	-0.019 [-0.033, -0.00723]
FD002 hybrid test ECE5	156/259	0.0477 [0.0302, 0.0682]	0.0411 [0.025, 0.0603]	-0.00665 [-0.00996, -0.00368]
FD002 hybrid test ECE10	156/259	0.0472 [0.0335, 0.0687]	0.0419 [0.0297, 0.0622]	-0.00527 [-0.009, -0.00178]
FD002 hybrid test incompatibility rate	156/259	8.9% [6.6%, 11.4%]	0.0% [0.0%, 0.0%]	-8.9% [-11.4%, -6.6%]
FD002 hybrid calibration Brier	156/52	0.0236 [0.00635, 0.0435]	0.0176 [0.00452, 0.0348]	-0.00594 [-0.0142, 0.000826]
FD002 hybrid calibration NLL	156/52	0.0795 [0.026, 0.141]	0.059 [0.0207, 0.108]	-0.0205 [-0.0493, 0.00259]
FD002 hybrid calibration ECE5	156/52	0.0426 [0.0141, 0.0776]	0.0333 [0.0131, 0.0664]	-0.00928 [-0.0169, 0.000671]
FD002 hybrid calibration ECE10	156/52	0.0474 [0.0187, 0.0819]	0.0431 [0.0173, 0.0761]	-0.00428 [-0.0125, 0.00321]
FD002 hybrid calibration incompatibility rate	156/52	8.5% [3.8%, 13.8%]	0.0% [0.0%, 0.0%]	-8.5% [-13.8%, -3.8%]
FD003 lstm test Brier	60/100	0.0267 [0.00832, 0.0519]	0.026 [0.0104, 0.0469]	-0.000682 [-0.00803, 0.00643]
FD003 lstm test NLL	60/100	0.0935 [0.0361, 0.174]	0.0879 [0.0402, 0.153]	-0.00563 [-0.0282, 0.0131]
FD003 lstm test ECE5	60/100	0.0299 [0.0143, 0.0612]	0.0163 [0.0105, 0.0458]	-0.0136 [-0.0194, 0.00167]
FD003 lstm test ECE10	60/100	0.0382 [0.0231, 0.0704]	0.0261 [0.0197, 0.0543]	-0.0122 [-0.0202, 0.00503]
FD003 lstm test incompatibility rate	60/100	12.8% [8.2%, 18.0%]	0.0% [0.0%, 0.0%]	-12.8% [-18.0%, -8.2%]
FD003 lstm calibration Brier	60/20	0.0503 [7.98e-08, 0.139]	0.0403 [7.98e-08, 0.111]	-0.01 [-0.0284, 0.000173]
FD003 lstm calibration NLL	60/20	0.143 [7.74e-05, 0.393]	0.119 [7.74e-05, 0.324]	-0.0234 [-0.0691, 0.00266]

Group / role / metric	Fit / evaluation engines	Raw [CI]	Projected [CI]	Paired difference [CI]
FD003 lstm calibration ECE5	60/20	0.0666 [7.74e-05, 0.178]	0.0575 [7.74e-05, 0.154]	-0.00917 [-0.0233, 0.00257]
FD003 lstm calibration ECE10	60/20	0.0671 [7.74e-05, 0.178]	0.0586 [7.74e-05, 0.155]	-0.00844 [-0.0225, 0.00257]
FD003 lstm calibration incompatibility rate	60/20	8.0% [0.0%, 18.0%]	0.0% [0.0%, 0.0%]	-8.0% [-18.0%, 0.0%]
FD003 hybrid test Brier	60/100	0.0291 [0.0118, 0.0515]	0.0227 [0.0095, 0.0409]	-0.00643 [-0.0129, -0.000557]
FD003 hybrid test NLL	60/100	0.0933 [0.0436, 0.159]	0.0767 [0.0362, 0.132]	-0.0166 [-0.0326, -0.00188]
FD003 hybrid test ECE5	60/100	0.0425 [0.0219, 0.0722]	0.0284 [0.0181, 0.0543]	-0.0141 [-0.0214, -0.000427]
FD003 hybrid test ECE10	60/100	0.047 [0.0283, 0.0777]	0.0361 [0.0241, 0.0606]	-0.011 [-0.0205, 0.00123]
FD003 hybrid test incompatibility rate	60/100	11.8% [7.6%, 16.4%]	0.0% [0.0%, 0.0%]	-11.8% [-16.4%, -7.6%]
FD003 hybrid calibration Brier	60/20	0.0827 [5.85e-05, 0.208]	0.066 [5.85e-05, 0.165]	-0.0167 [-0.0438, 0]
FD003 hybrid calibration NLL	60/20	0.295 [0.00212, 0.759]	0.237 [0.00212, 0.607]	-0.0578 [-0.152, 0]
FD003 hybrid calibration ECE5	60/20	0.0895 [0.00209, 0.226]	0.0777 [0.00209, 0.195]	-0.0118 [-0.0311, 0]
FD003 hybrid calibration ECE10	60/20	0.09 [0.00209, 0.226]	0.0786 [0.00209, 0.196]	-0.0114 [-0.0307, 0]
FD003 hybrid calibration incompatibility rate	60/20	5.0% [0.0%, 14.0%]	0.0% [0.0%, 0.0%]	-5.0% [-14.0%, 0.0%]
FD004 lstm test Brier	149/248	0.0529 [0.0339, 0.0746]	0.0457 [0.0289, 0.0651]	-0.00721 [-0.0124, -0.00267]
FD004 lstm test NLL	149/248	0.173 [0.108, 0.249]	0.146 [0.0939, 0.207]	-0.0267 [-0.0486, -0.00783]
FD004 lstm test ECE5	149/248	0.0563 [0.0323, 0.0844]	0.0426 [0.0198, 0.0689]	-0.0137 [-0.0179, -0.00919]
FD004 lstm test ECE10	149/248	0.0593 [0.0405, 0.0886]	0.0459 [0.0292, 0.073]	-0.0134 [-0.019, -0.00907]
FD004 lstm test incompatibility rate	149/248	16.6% [13.1%, 20.3%]	0.0% [0.0%, 0.0%]	-16.6% [-20.3%, -13.1%]
FD004 lstm calibration Brier	149/50	0.0143 [0.00139, 0.033]	0.00866 [0.00056, 0.0195]	-0.00569 [-0.0135, -0.000315]
FD004 lstm calibration NLL	149/50	0.0472 [0.011, 0.0985]	0.0335 [0.00816, 0.0679]	-0.0137 [-0.0316, -0.000622]
FD004 lstm calibration ECE5	149/50	0.0314 [0.00797, 0.0652]	0.0218 [0.00483, 0.0481]	-0.00958 [-0.0173, -0.00217]
FD004 lstm calibration ECE10	149/50	0.0337 [0.00995, 0.0666]	0.0273 [0.00793, 0.0536]	-0.00637 [-0.0145, -2.57e-05]
FD004 lstm calibration incompatibility rate	149/50	12.0% [5.2%, 20.0%]	0.0% [0.0%, 0.0%]	-12.0% [-20.0%, -5.2%]
FD004 hybrid test Brier	149/248	0.0448 [0.0279, 0.0642]	0.0403 [0.0248, 0.0576]	-0.00456 [-0.01, -0.000215]
FD004 hybrid test NLL	149/248	0.146 [0.0912, 0.208]	0.128 [0.0823, 0.179]	-0.0186 [-0.0389, -0.00249]
FD004 hybrid test ECE5	149/248	0.0379 [0.0177, 0.065]	0.0295 [0.0114, 0.0546]	-0.0084 [-0.0128, -0.00201]
FD004 hybrid test ECE10	149/248	0.0462 [0.0336, 0.0742]	0.0381 [0.0285, 0.0643]	-0.00802 [-0.0144, -0.00162]
FD004 hybrid test incompatibility rate	149/248	12.7% [9.8%, 15.6%]	0.0% [0.0%, 0.0%]	-12.7% [-15.6%, -9.8%]
FD004 hybrid calibration Brier	149/50	0.0191 [0.00198, 0.048]	0.0145 [0.00179, 0.0329]	-0.0046 [-0.0158, 0.00213]
FD004 hybrid calibration NLL	149/50	0.0625 [0.0129, 0.143]	0.0504 [0.0129, 0.103]	-0.0122 [-0.0408, 0.00517]
FD004 hybrid calibration ECE5	149/50	0.035 [0.00719, 0.0757]	0.026 [0.00703, 0.0593]	-0.00901 [-0.0186, 0.00119]

Group / role / metric	Fit / evaluation engines	Raw [CI]	Projected [CI]	Paired difference [CI]
FD004 hybrid calibration ECE10	149/50	0.0387 [0.0119, 0.0782]	0.0318 [0.0111, 0.0644]	-0.00682 [-0.0169, 0.00234]
FD004 hybrid calibration incompatibility rate	149/50	10.0% [4.4%, 16.4%]	0.0% [0.0%, 0.0%]	-10.0% [-16.4%, -4.4%]
FD001 combined test Brier	60/100	0.0352 [0.0143, 0.0617]	0.0348 [0.0151, 0.0603]	-0.000343 [-0.00646, 0.00523]
FD001 combined test NLL	60/100	0.122 [0.0557, 0.208]	0.117 [0.0536, 0.201]	-0.00477 [-0.0204, 0.00936]
FD001 combined test ECE5	60/100	0.0163 [0.0121, 0.0504]	0.0133 [0.0117, 0.0464]	-0.00307 [-0.0175, 0.0157]
FD001 combined test ECE10	60/100	0.0387 [0.0312, 0.0745]	0.028 [0.0224, 0.061]	-0.0106 [-0.025, 0.00149]
FD001 combined test incompatibility rate	60/100	17.0% [11.8%, 22.4%]	0.0% [0.0%, 0.0%]	-17.0% [-22.4%, -11.8%]
FD001 combined calibration Brier	60/20	0.0405 [0.000109, 0.119]	0.0389 [7.96e-05, 0.116]	-0.00162 [-0.00406, 0]
FD001 combined calibration NLL	60/20	0.127 [0.00255, 0.359]	0.119 [0.00205, 0.348]	-0.00747 [-0.019, 0]
FD001 combined calibration ECE5	60/20	0.0524 [0.00249, 0.146]	0.0464 [0.00107, 0.137]	-0.00605 [-0.0164, 0.000459]
FD001 combined calibration ECE10	60/20	0.0534 [0.00249, 0.147]	0.0472 [0.00201, 0.137]	-0.00623 [-0.0164, 8.65e-05]
FD001 combined calibration incompatibility rate	60/20	8.5% [0.0%, 18.0%]	0.0% [0.0%, 0.0%]	-8.5% [-18.0%, 0.0%]
FD002 combined test Brier	156/259	0.0292 [0.0162, 0.044]	0.0248 [0.0137, 0.0378]	-0.00444 [-0.0072, -0.00199]
FD002 combined test NLL	156/259	0.0999 [0.0572, 0.15]	0.085 [0.049, 0.128]	-0.0149 [-0.024, -0.00663]
FD002 combined test ECE5	156/259	0.0418 [0.0254, 0.0618]	0.0356 [0.0201, 0.0535]	-0.00622 [-0.00913, -0.00415]
FD002 combined test ECE10	156/259	0.0418 [0.0283, 0.0618]	0.0362 [0.0245, 0.0549]	-0.00562 [-0.00809, -0.00299]
FD002 combined test incompatibility rate	156/259	9.2% [6.9%, 11.8%]	0.0% [0.0%, 0.0%]	-9.2% [-11.8%, -6.9%]
FD002 combined calibration Brier	156/52	0.016 [0.00394, 0.0314]	0.0124 [0.00244, 0.0261]	-0.00356 [-0.00786, 1.29e-05]
FD002 combined calibration NLL	156/52	0.0562 [0.0187, 0.102]	0.0441 [0.0141, 0.0835]	-0.0121 [-0.0279, 0.000456]
FD002 combined calibration ECE5	156/52	0.0332 [0.00998, 0.0634]	0.0262 [0.00942, 0.0547]	-0.00699 [-0.012, -0.000139]
FD002 combined calibration ECE10	156/52	0.0371 [0.0136, 0.0666]	0.0336 [0.0122, 0.0615]	-0.0035 [-0.00917, 0.00148]
FD002 combined calibration incompatibility rate	156/52	9.6% [4.4%, 15.6%]	0.0% [0.0%, 0.0%]	-9.6% [-15.6%, -4.4%]
FD003 combined test Brier	60/100	0.0279 [0.0108, 0.0512]	0.0243 [0.0103, 0.043]	-0.00356 [-0.00974, 0.00238]
FD003 combined test NLL	60/100	0.0934 [0.0411, 0.164]	0.0823 [0.0387, 0.14]	-0.0111 [-0.0287, 0.00455]
FD003 combined test ECE5	60/100	0.0362 [0.0196, 0.0662]	0.0223 [0.0157, 0.049]	-0.0138 [-0.0193, -0.000698]
FD003 combined test ECE10	60/100	0.0426 [0.0265, 0.0728]	0.0311 [0.0225, 0.0567]	-0.0116 [-0.0189, 0.00223]
FD003 combined test incompatibility rate	60/100	12.3% [8.4%, 16.6%]	0.0% [0.0%, 0.0%]	-12.3% [-16.6%, -8.4%]
FD003 combined calibration Brier	60/20	0.0665 [2.92e-05, 0.174]	0.0531 [2.92e-05, 0.138]	-0.0134 [-0.0361, 8.65e-05]
FD003 combined calibration NLL	60/20	0.219 [0.00108, 0.576]	0.178 [0.00108, 0.464]	-0.0406 [-0.111, 0.00133]

Group / role / metric	Fit / evaluation engines	Raw [CI]	Projected [CI]	Paired difference [CI]
FD003 combined calibration ECE5	60/20	0.0781 [0.00106, 0.202]	0.0676 [0.00106, 0.175]	-0.0105 [-0.0272, 0.00129]
FD003 combined calibration ECE10	60/20	0.0785 [0.00106, 0.202]	0.0686 [0.00106, 0.176]	-0.00991 [-0.0266, 0.00129]
FD003 combined calibration incompatibility rate	60/20	6.5% [0.0%, 14.0%]	0.0% [0.0%, 0.0%]	-6.5% [-14.0%, 0.0%]
FD004 combined test Brier	149/248	0.0489 [0.0314, 0.0692]	0.043 [0.0271, 0.0614]	-0.00588 [-0.0105, -0.00205]
FD004 combined test NLL	149/248	0.159 [0.1, 0.227]	0.137 [0.0891, 0.194]	-0.0226 [-0.0407, -0.00728]
FD004 combined test ECE5	149/248	0.0471 [0.0252, 0.0744]	0.036 [0.0161, 0.0615]	-0.0111 [-0.0149, -0.00629]
FD004 combined test ECE10	149/248	0.0527 [0.0378, 0.0812]	0.042 [0.0298, 0.0683]	-0.0107 [-0.0158, -0.0062]
FD004 combined test incompatibility rate	149/248	14.6% [11.8%, 17.8%]	0.0% [0.0%, 0.0%]	-14.6% [-17.8%, -11.8%]
FD004 combined calibration Brier	149/50	0.0167 [0.00221, 0.04]	0.0116 [0.00181, 0.0257]	-0.00514 [-0.0141, 0.000347]
FD004 combined calibration NLL	149/50	0.0549 [0.0136, 0.118]	0.042 [0.0118, 0.0835]	-0.0129 [-0.0351, 0.000975]
FD004 combined calibration ECE5	149/50	0.0332 [0.0081, 0.0688]	0.0239 [0.00643, 0.0535]	-0.0093 [-0.0173, -0.000832]
FD004 combined calibration ECE10	149/50	0.0362 [0.012, 0.0718]	0.0296 [0.0102, 0.0583]	-0.00659 [-0.0148, 0.000369]
FD004 combined calibration incompatibility rate	149/50	11.0% [5.4%, 17.4%]	0.0% [0.0%, 0.0%]	-11.0% [-17.4%, -5.4%]
FD001-FD004 combined test Brier	425/707	0.0368 [0.0276, 0.0467]	0.0325 [0.0244, 0.0414]	-0.00424 [-0.00647, -0.00226]
FD001-FD004 combined test NLL	425/707	0.123 [0.0917, 0.155]	0.107 [0.0814, 0.135]	-0.0156 [-0.0236, -0.00874]
FD001-FD004 combined test ECE5	425/707	0.0393 [0.0301, 0.0535]	0.0307 [0.0237, 0.0444]	-0.00855 [-0.0109, -0.00443]
FD001-FD004 combined test ECE10	425/707	0.0453 [0.0393, 0.0612]	0.0363 [0.0319, 0.0507]	-0.00896 [-0.0118, -0.00584]
FD001-FD004 combined test incompatibility rate	425/707	12.7% [11.1%, 14.4%]	0.0% [0.0%, 0.0%]	-12.7% [-14.4%, -11.1%]
FD001-FD004 combined calibration Brier	425/142	0.0268 [0.0109, 0.0465]	0.0216 [0.00792, 0.0387]	-0.00522 [-0.00964, -0.00163]
FD001-FD004 combined calibration NLL	425/142	0.0886 [0.0387, 0.149]	0.0728 [0.0304, 0.125]	-0.0158 [-0.0286, -0.00501]
FD001-FD004 combined calibration ECE5	425/142	0.0422 [0.0205, 0.0678]	0.0341 [0.0159, 0.0578]	-0.00817 [-0.0119, -0.00358]
FD001-FD004 combined calibration ECE10	425/142	0.0449 [0.0238, 0.07]	0.039 [0.0206, 0.062]	-0.00588 [-0.00992, -0.00192]
FD001-FD004 combined calibration incompatibility rate	425/142	9.5% [6.3%, 13.0%]	0.0% [0.0%, 0.0%]	-9.5% [-13.0%, -6.3%]

Table S14. All adjusted effects, relative upper endpoints and approximate MDEs

Source: `C_effect_sizes_power.csv`. Each ratio uses its outcome's observed mean MSE. Pooled estimates are engine-count-weighted averages of group-specific effects; normal/Wald MDEs are approximate.

Group / outcome	Fit / test engines	Coefficient [CI]	Mean MSE	Upper CI / mean	MDE (MDE / mean)
FD001 lstm median squared error	60/100	69.7 [-167, 405]	217	187.0%	429 (198.1%)
FD001 lstm Brier	60/100	0.0426 [-0.0369, 0.124]	0.0356	348.9%	0.115 (324.3%)

Group / outcome	Fit / test engines	Coefficient [CI]	Mean MSE	Upper CI / mean	MDE (MDE / mean)
FD001 hybrid median squared error	60/100	-250 [-415, -95.3]	259	-36.8%	227 (87.6%)
FD001 hybrid Brier	60/100	-0.0143 [-0.0887, 0.055]	0.0348	158.1%	0.105 (301.0%)
FD002 lstm median squared error	156/259	-138 [-240, -51]	205	-24.8%	136 (66.3%)
FD002 lstm Brier	156/259	0.0908 [0.0332, 0.155]	0.0251	615.7%	0.0889 (354.4%)
FD002 hybrid median squared error	156/259	-149 [-311, -4.99]	241	-2.1%	214 (88.7%)
FD002 hybrid Brier	156/259	0.103 [0.0365, 0.171]	0.0333	513.2%	0.0945 (283.6%)
FD003 lstm median squared error	60/100	-320 [-595, -71]	323	-22.0%	380 (117.6%)
FD003 lstm Brier	60/100	0.0822 [0.0155, 0.179]	0.0267	672.7%	0.12 (450.5%)
FD003 hybrid median squared error	60/100	-99.9 [-230, 29]	234	12.4%	186 (79.3%)
FD003 hybrid Brier	60/100	0.129 [0.0524, 0.209]	0.0291	716.3%	0.113 (387.4%)
FD004 lstm median squared error	149/248	-36.3 [-276, 258]	275	93.8%	384 (139.6%)
FD004 lstm Brier	149/248	0.124 [0.0554, 0.202]	0.0529	382.0%	0.106 (200.6%)
FD004 hybrid median squared error	149/248	90.2 [-147, 429]	274	156.5%	430 (157.0%)
FD004 hybrid Brier	149/248	0.103 [0.0247, 0.187]	0.0448	416.7%	0.117 (260.6%)
FD001 combined median squared error	60/100	-90.2 [-271, 118]	238	49.7%	285 (120.0%)
FD001 combined Brier	60/100	0.0141 [-0.0532, 0.0796]	0.0352	226.2%	0.097 (275.8%)
FD002 combined median squared error	156/259	-144 [-270, -34.6]	223	-15.5%	167 (74.8%)
FD002 combined Brier	156/259	0.0971 [0.0447, 0.148]	0.0292	506.2%	0.0745 (254.9%)
FD003 combined median squared error	60/100	-210 [-390, -42.2]	279	-15.1%	254 (91.1%)
FD003 combined Brier	60/100	0.106 [0.0437, 0.18]	0.0279	644.3%	0.0967 (346.6%)
FD004 combined median squared error	149/248	27 [-204, 343]	274	125.0%	401 (146.2%)
FD004 combined Brier	149/248	0.113 [0.049, 0.186]	0.0489	381.0%	0.0965 (197.5%)
FD001-FD004 combined median squared error	425/707	-85.6 [-191, 37.1]	251	14.8%	162 (64.4%)
FD001-FD004 combined Brier	425/707	0.0922 [0.0589, 0.126]	0.0368	343.7%	0.0485 (131.9%)